🔬 Overview over PT performance data and settings

- This page contains an overview over the key information related to all pulse tubes
- The components listed below are the ones shipped to Denver. (Information on the replacement of RVs or VHs during the optimization process (if required) can be found in the lab journal.
- Valve settings in rotations are counted relative to fully closed valve
- Valve settings in mm are measured with a caliper from the end of the needle valve to the rim on the socket

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1. Overview over systems for single compressor operation

(standard data points for P1=0W, 20W and P2=0W, 0.5W)

SN	T1 @ 20W / 0.5W	T2 @ 20W / 0.5W	T1 @ 20W / 0W	T2 @ 20W / 0W	T1 @ 0W / 0.5W	T2 @ 0W / 0.5W	T1 @ 0W / 0W	T2 @ 0W / 0W	# of Compressors	Type of Compr.	f / Hz
S0007	62.7	3.64	62.7	2.36	45.2	3.69	44.2	2.28	1x	F-70	1.0
S0008	62.2	3.47	61.4	2.51	44.5	3.47	42.8	2.50	1x	F-70	1.0
S0009	53.61	3.45	52.09	2.39	39.51	3.45	37.90	2.38	1x	F70	1.0
S0016 (IBM)	59.74	3.37	58.89	2.32	43.86	3.36	42.07	2.25	1x	F-70	1.0
S0018 (IBM)	49.8	3.4	48.0	2.48	35.9	3.29	34.9	2.27	1x	F-70	1.0
S0019	61.5	3.59	61.22	2.52	45.8	3.59	44.8	2.48	1x	F-70	1.1
S0022	58.1	3.54	57.6	2.39	43.6	3.56	39.46	2.44	1x	F-70	1.1
S0023	60.88	3.55	60.29	2.48	45.99	3.53	42.85	2.44	1 x	F-70	1.1
S0028	58.03	3.38	57.70	2.30	43.38	3.44	42.66	2.27	1x	F-70	1.0
	not shipped with single compressor tuning but fully characterized										
S0034	59.01	3.64	59.33	2.39	43.12	3.66	39.84	2.29	1x	F-70	0.9
S0038	new acceptance criteria, standard loadmap-points changed (20W → 25W)								1x	F70	1.0
S0037	55.81	3.49	55.40	2.12	40.98	3.60	40.34	2.07	1x	F70	1.1

S0035	new acceptance criteria, standard loadmap-points changed (20W → 25W)								1x	F70	1.1
S0048	57.58	3.59	56.94	2.39	43.51	3.62	41.94	2.29	1x	F70	1.1
S0049	53.87	3.42	53.36	2.14	42.25	3.50	41.46	2.09	1x	F70	1.1

2. Overview over systems for double compressor operation

(standard data points for P1=0W, 20W and P2=0W, 0.5W)

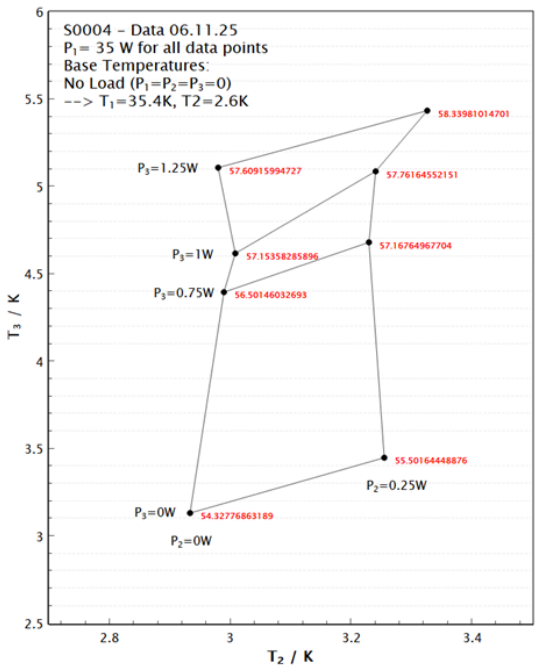
SN	T1 @ 20W / 0.5W	T2 @ 20W / 0.5W	T1 @ 20W / 0W	T2 @ 20W / 0W	T1 @ 0W / 0.5W	T2 @ 0W / 0.5W	T1 @ 0W / 0W	T2 @ 0W / 0W	# of Compressors	Type of Compr.	f / Hz
S0004	IBM, no fully standard load-maps taken								2x	F-70	1.1
S0005	IBM, no fully standard load-maps taken								2x	F-70	1.1
S0006	IBM, no fully standard load-maps taken								2x	F-70	1.1
S0010	points here do not align with loadmap to document spec requirement								2x	F-70	1.1
S00017	IBM, no fully standard load-maps taken								2x	F-70	1.1
S0028	49.95	3.12	48.15	2.56	37.28	3.13	36.45	2.47	2x	F-70	1.1

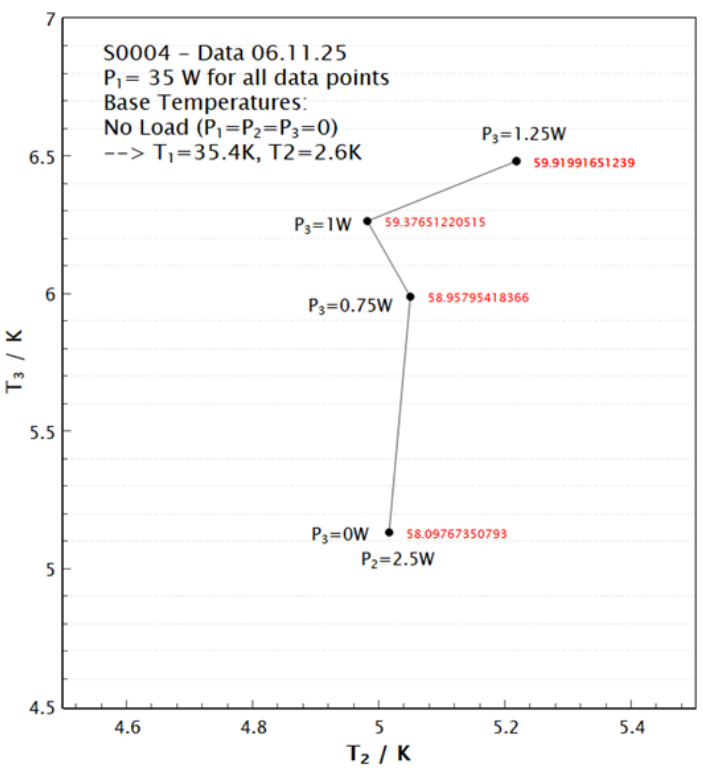
3. Details on Cooler Performance

S0004 - IBM PT-build #1 - 2 x F70 (shipped)

Key Information on S0004	
Cold Foot	CF0004

Valve head	VH0001	
Rotary Valve	RV0005	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	Nov. 6th, 2025	
Shipping date	Nov. 10th, 2025	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 9.375 DIV1A: 8 DIV1B: 3 ORV2: 8 DIV2A: 8.25 DIV2B: 9.75 f / Hz: 1.1	Protrusion (of fine metering valve screw from socket) ORV1: 11.18 mm DIV1A: 9.85 mm DIV1B: 7.35 mm ORV2: 10.27 mm DIV2A: 11.09 mm DIV2B: 11.25 mm f / Hz: 1.1
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • operated using 2 Compressors (2xF70H)	
Load-Maps and Performance of S0004		

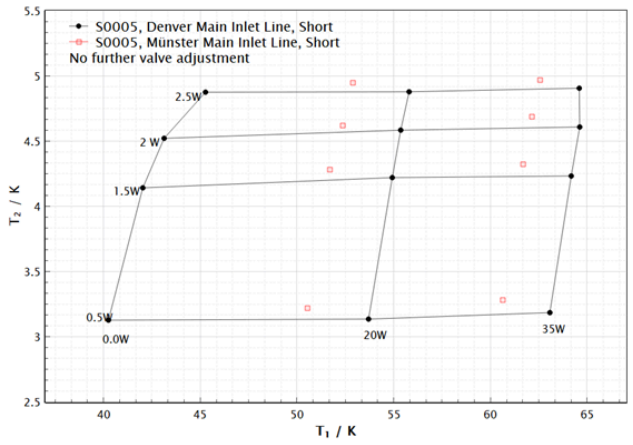
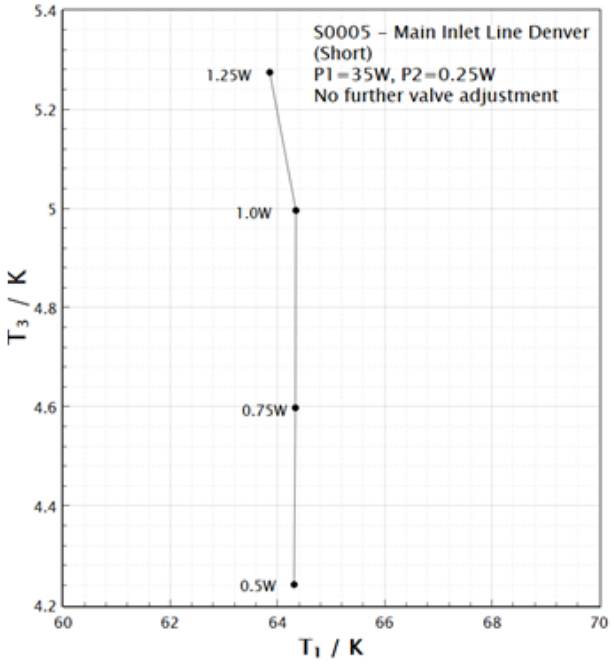




Key Points of the Load Map	not applicable for IBM system
Acceptance criteria	not applicable for IBM system
Control limits	not applicable for IBM system

S0005 - IBM PT-build #2 - 2 x F70 (shipped)

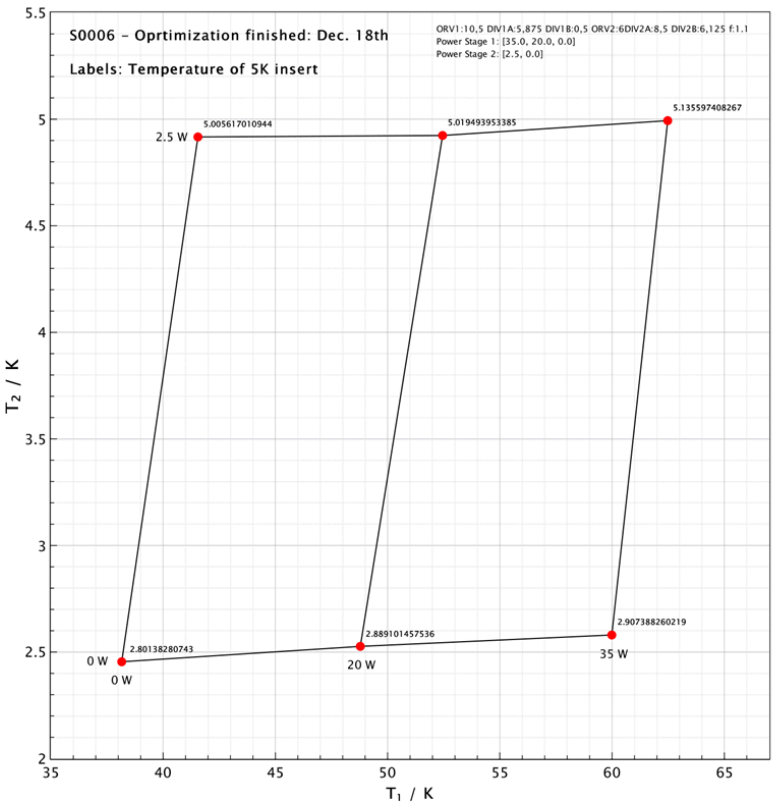
Key Information on S0005		
Cold Foot	CF0005	
Valve head	VH0002	
Rotary Valve	RV0004	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	Nov. 4th, 2025	
Shipping date	Nov. 10th, 2025	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 8.25	Protrusion (of fine metering valve screw from socket) ORV1: 10.23 mm

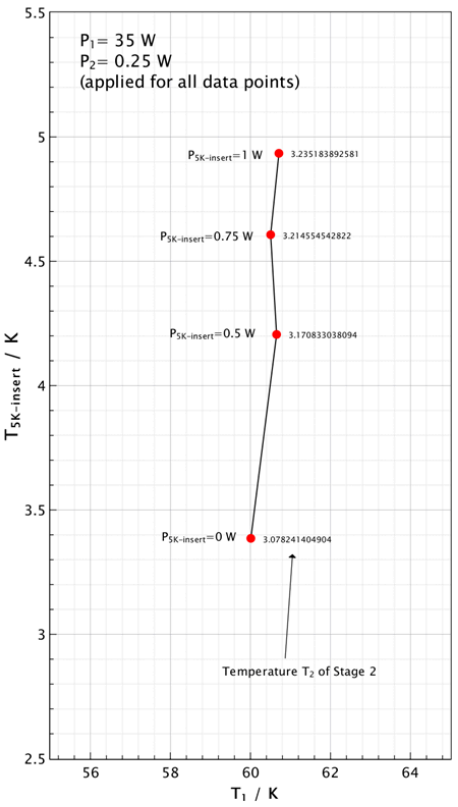
	DIV1A: 5.5 DIV1B: 3.0 ORV2: 5.0 DIV2A: 6.25 DIV2B: 8.875 f / Hz: 1.1	DIV1A: 8.55 mm DIV1B: 6.86 mm ORV2: 8.25 mm DIV2A: 9.06 mm DIV2B: 11.0 mm f / Hz: 1.1
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • operated using 2 Compressors (2xF70H)	
Load-Maps and Performance of S0005		
		
Key Points of the Load Map	not applicable for IBM system	
Acceptance criteria	not applicable for IBM system	
Control limits	not applicable for IBM system	

S0006 - IBM PT-build #3 - 2 x F70 (shipped)

Key Information on S0006	
Cold Foot	CF0006

Valve head	VH0006	
Rotary Valve	RV0005	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	Dec. 19th, 2025	
Shipping date	Jan. 16th, 2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.5 DIV1A: 5.875 DIV1B: 1 ORV2: 6 DIV2A: 8.5 DIV2B: 6.125 f / Hz: 1.1	Protrusion (of fine metering valve screw from socket) ORV1: 11.75 mm DIV1A: 8.76 mm DIV1B: 5.65 mm ORV2: 8.99 mm DIV2A: 10.35 mm DIV2B: 8.77 mm f / Hz: 1.1
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • operated using 2 Compressors (2xF70H)	
Load-Maps and Performance of S0006		





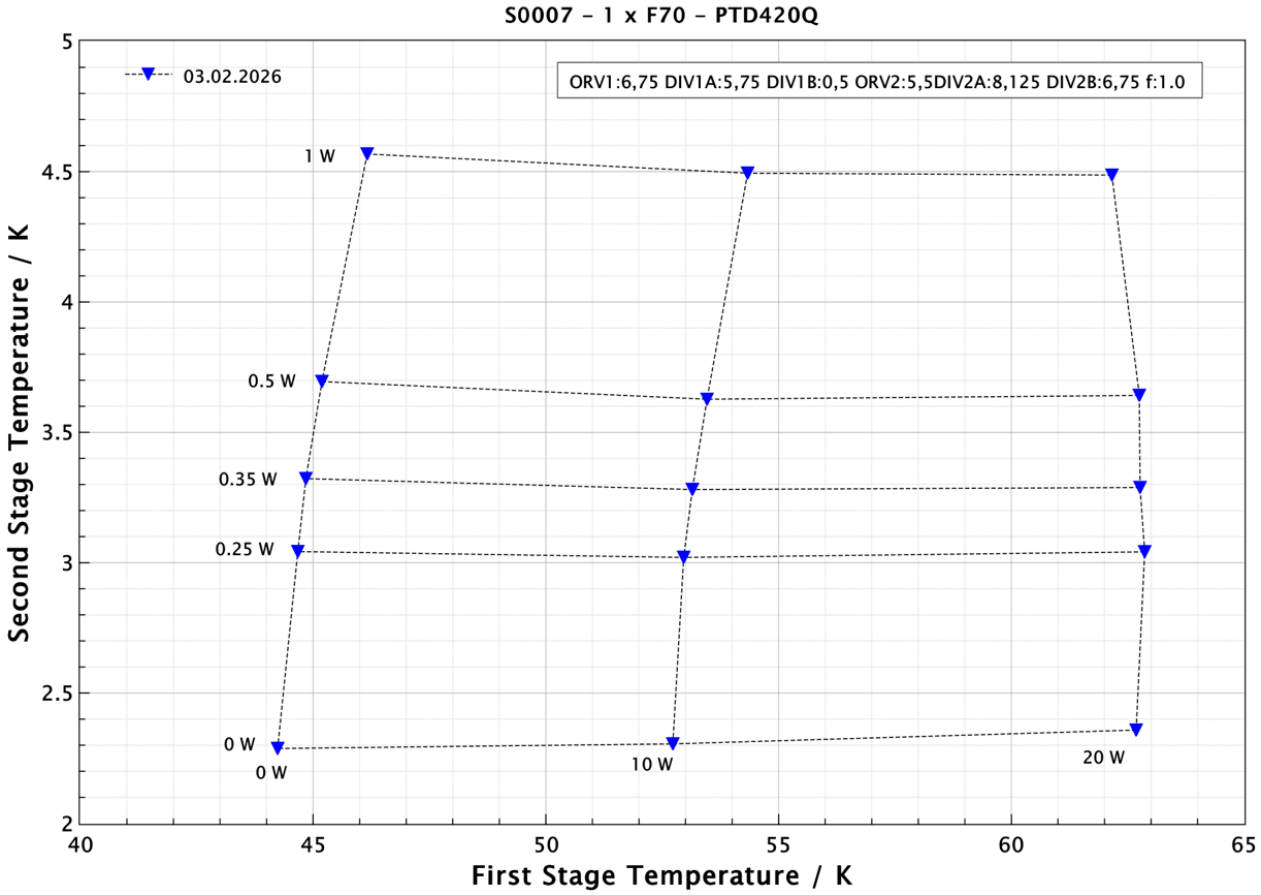
Key Points of the Load Map	not applicable for IBM system
Acceptance criteria	not applicable for IBM system
Control limits	not applicable for IBM system

S0007 - regular 1.5 - single F70 (shipped)

Key Information on S0007		
Cold Foot	CF0007	
Valve head	VH0007	
Rotary Valve	RV0002	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	Feb. 3rd, 2026	
Shipping date	shipping date Feb. 27th, 2026	
Final valve settings:	Rotations (counted relative to fully closed valve)	Protrusion (of fine metering valve screw from socket)

	ORV1: 6.75 DIV1A: 5.75 DIV1B: 0.5 ORV2: 5.5 DIV2A: 8.125 DIV2B: 6.75 f / Hz: 1.0	ORV1: 9.46 mm DIV1A: 4.77 mm DIV1B: 7.79 mm ORV2: 8.03 mm DIV2A: 10.04 mm DIV2B: 9.29 mm f / Hz: 1.0
Comments / remarks	• MQI-PT420-aq v1.5 • Optimized for a single F70 compressor	

Load-Maps and Performance of S0007

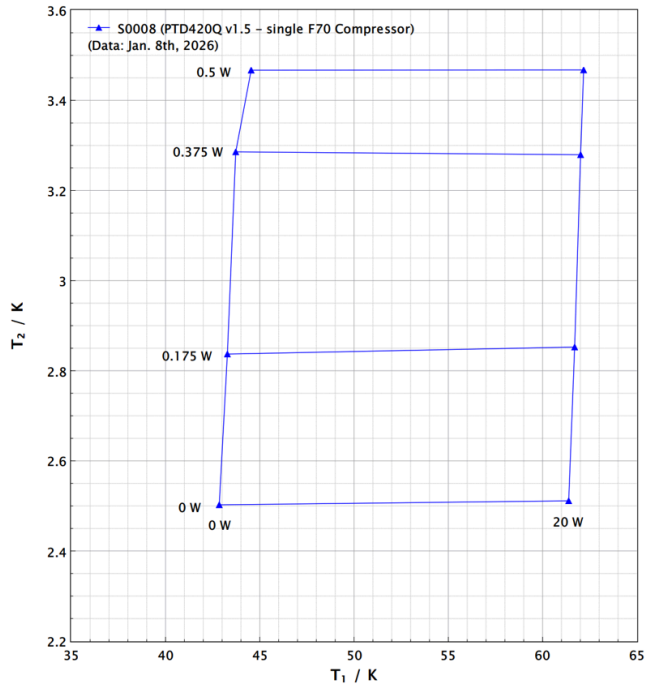


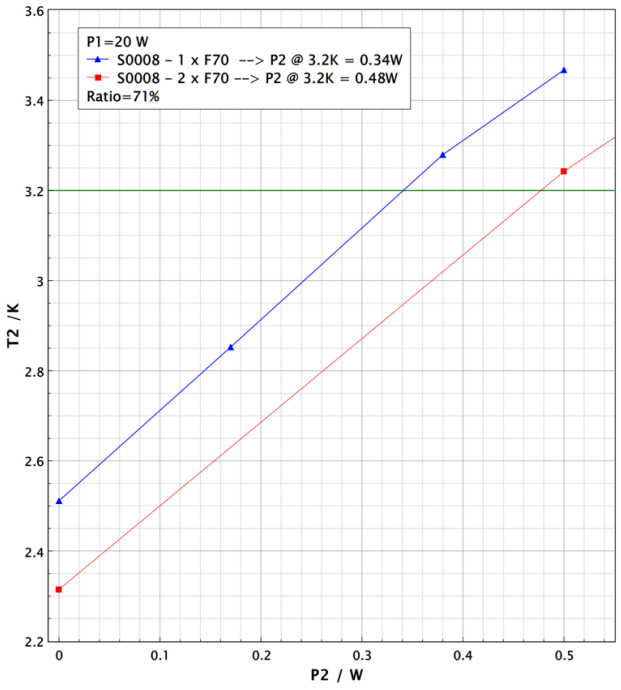
Key Points of the Load Map	P1,	P2,	T1,	T2
	0	0	44.2	2.29
	0	0.5	45.2	3.69
	20	0	62.7	2.36
	20	0.5	62.7	3.64

Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined yet

S0008 - regular 1.5 - single F70 (shipped)

Key Information on S0008		
Cold Foot	CF0008	
Valve head	VH0008	
Rotary Valve	RV0008	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	Jan. 8th, 2026	
Shipping date	Jan. 16th, 2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 6.75 DIV1A: 5.75 DIV1B: 0.5 ORV2: 6.125 DIV2A: 5.675 DIV2B: 8.25 f / Hz: 1.0	Protrusion (of fine metering valve screw from socket) ORV1: 9.72 mm DIV1A: 8.64 mm DIV1B: 5.23 mm ORV2: 8.75 mm DIV2A: 8.73 mm DIV2B: 10.28 mm f / Hz: 1.0
Comments / remarks	• MQI-PT420-aq v1.5 • Optimized for a single compressor	
Load-Maps and Performance of S0008		





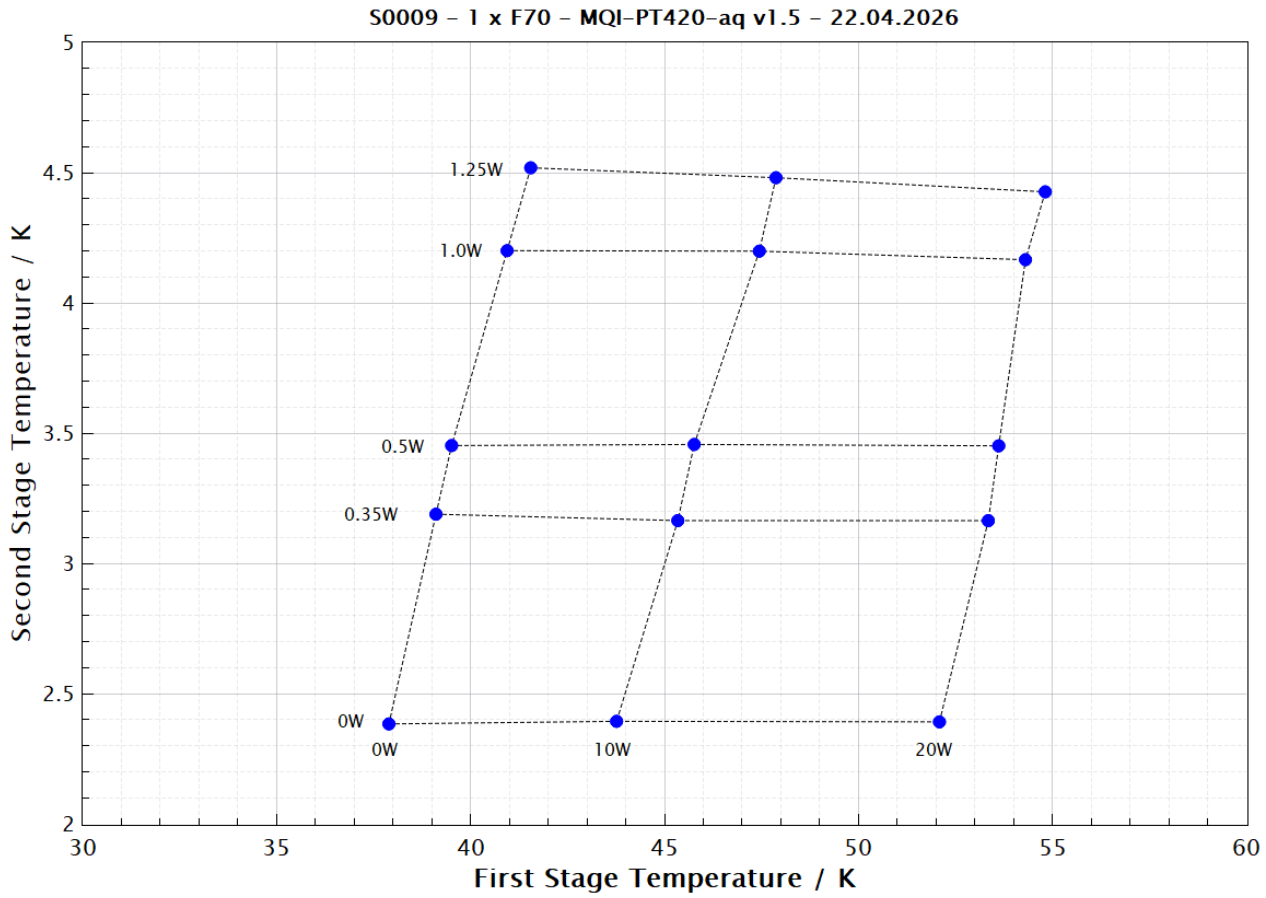
Key Points of the Load Map	P1, P2, T1, T2 0, 0, 42.8, 2.50 20, 0, 61.4, 2.51 0, 0.5, 44.5, 3.47 20, 0.5, 62.2, 3.47
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping

S0009 - regular 1.5 - single F70 (shipped)

Key Information on S0009	
Cold Foot	CF0009
Valve head	VH0005
Rotary Valve	RV0010
Main Inlet Line	Nr. 400-000264
Final load maps recorded	April, 22nd 2026
Shipping date	April, 30th 2026

Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 9.375 DIV1A: 5.75 DIV1B: 0.5 ORV2: 5.5 DIV2A: 8.5 DIV2B: 6.75 f / Hz: 1.0	Protrusion (of fine metering valve screw from socket) ORV1: 11.25 DIV1A: 8.79 DIV1B: 5.57 ORV2: 8.51 DIV2A: 10.40 DIV2B: 9.58 f=1.0 Hz
Comments / remarks	• MQI-PT420-aq v1.5 • Optimized for a single compressor	

Load-Maps and Performance of S0009



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 37.8956947129077, 2.38332943421858

	20, 0, 52.0894871568597, 2.3911986916641 0, 0.5, 39.5139864704434, 3.45200936872492 20, 0.5, 53.6177173973522, 3.45103266752766
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping

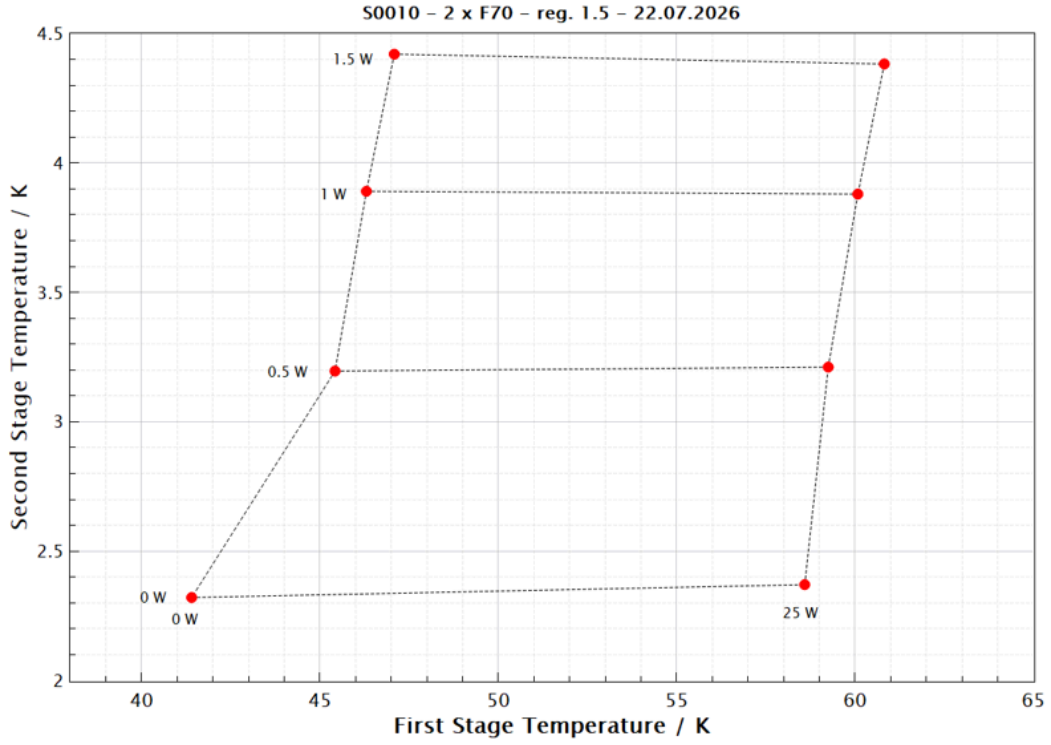
S0009 - Full Testing Configuration		
Component	**Number / Serial Number / Type**	**Comment**
Cold Foot	CF0009	
Rotary Valve	RV0007	
Valve Head	VH005	
Station	#2	
Main Inlet Line	Short, Isolation, Nr. 400-000264	
Flexlines	20m, DN25, Pair 2	
Buffer Volumes	2 x 5l, #1a and #1b	
Compressor 1	#4	
Compressor 2	-	
Vacuum Chamber	Chamber 1	
Lid for Vacuum Chamber	Lid #4	
Version of Software	V 56	
Initial filling pressure	17.1 bar	
Sensor T2	X237981	
Sensor T1	X254359	
Sensor T3	-	
Sensor PT2	PT100	

Heater Stage 2	100 Ohm	
Heater Stage 1	100 Ohm	
Heater Stage 3	-	
MLI Regenerator 2	7 Layers	
MLI IHE	-	
MLI Stage 2	5 Layers	
MLI Regenerator 1	7 Layers	
MLI Pulse Tube 1	7 Layers	
MLI Stage 1	4 Layers	

S0010 - regular 1.5 - 2 x F70 (not shipped until further notice)

Key Information on S0010	
Cold Foot	CF0010
Valve head	VH0034
Rotary Valve	RV0016 (used for final loadmap)
Main Inlet Line	900-0012322/1 (Medium length)
Final load maps recorded	22.07.2026
Shipping date	**Unit hits specs as a two compressor system, it remains in Germany for now as requested by Denver**

Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: DIV1A: DIV1B: ORV2: DIV2A: DIV2B: **f / Hz:**	Protrusion (of fine metering valve from socket) ORV1: 12.7 mm DIV1A: 8.65 mm DIV1B: 5.16 mm ORV2: 8.37 mm DIV2A: 10.68 mm DIV2B: 9.73 mm **f / Hz: 1.1**
Comments / remarks		
Load-Maps and Performance of S0010		



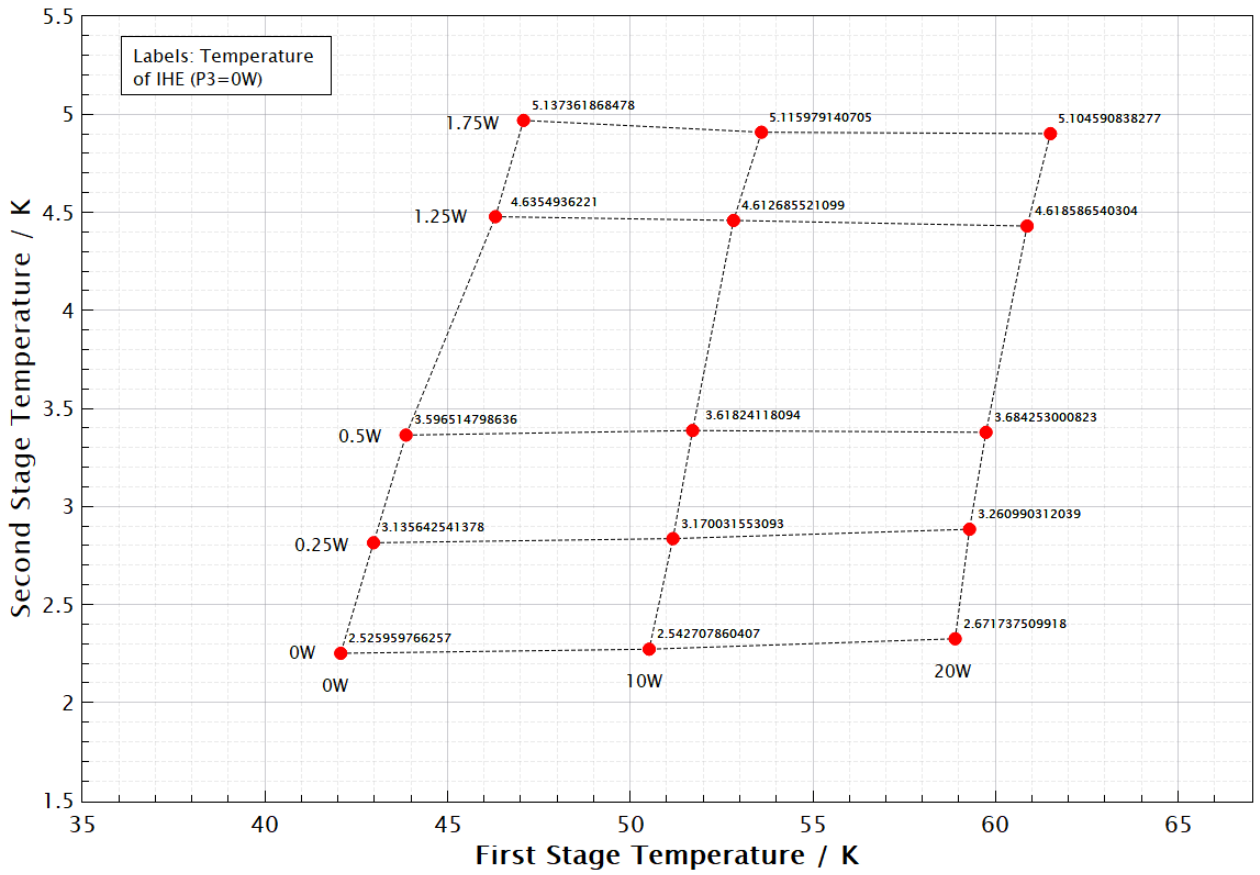
Key Points of the Load Map	P1, P2, T1, T2 0, 0, 41.4 K, 2.32 K 0, 0.5, 45.4 K, 3.19 K 20, 0, --, -- 20, 0.5, --, --
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping

S0010 - Full testing configuration		
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0010	
Rotary Valve	RV0016	
Valve Head	VH0034	
Measurement Station	#1	
Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#2	
Buffer Volumes	#2	
Compressor 1	F70 #1	2 compressor tuning
Compressor 2	F70 #2	
Vacuum Chamber	#5	
Lid for Vacuum Chamber	#3	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X254340	Plug4
Sensor T1	Cernox X254356	Plug 4
Sensor T3	-	
Sensor PT2	PT100	Plug 4
Heater Stage 2	100 Ohm	Plug 4
Heater Stage 1	80 Ohm	Plug 4
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	
MLI Stage 2	7 layers	
MLI Regenerator 1	7 layers	

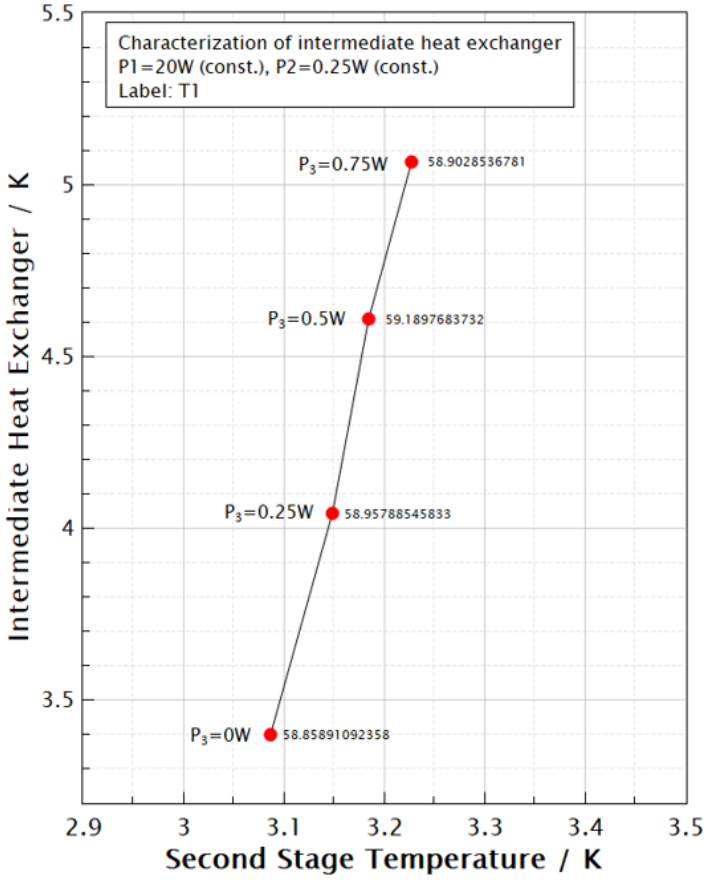
MLI Pulse Tube 1	10 layers	
MLI Stage 1	6 layers	

S0016 - IBM PT-build #4 - single F70 (shipped)

Key Information on S0016		
Cold Foot	CF0016	
Valve head	VH0028	
Rotary Valve	RV0011	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	March, 30th 2026	
Shipping date	April, 2nd 2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.5 DIV1A: 4 DIV1B: 0.5 ORV2: 4.25 DIV2A: 8.5 DIV2B:5.0 f / Hz: 1.0	Protrusion (of fine metering valve screw from socket) ORV1: 11.99 mm DIV1A: 7.81 mm DIV1B: 5.55 mm ORV2: 7.84 mm DIV2A: 10.57 mm DIV2B: 7.74 mm f / Hz: 1.0
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • **Optimized for a single compressor and IHE**	
Load-Maps and Performance of S0016		



Loadmap of S0016 using the revised rotary valve. A cooling power of appr. 21W @ 60K can be extracted with 0.25W applied to stage 1 (operation to drive the fridge)



Characterization of intermediate heat exchanger of S0016. Also with loads of 0.75W (IHE) and 0.25W (stage 2) a cooling power of 21W @ 60K can be extracted for the first stage

Key Points of the Load Map	P1, P2, T1, T2 0, 0, 42.07, 2.25 0, 0.5, 43.86, 3.36 20, 0, 58.89, 2.32 20, 0.5, 59.74, 3.37
Acceptance criteria	not applicable for IBM
Control limits	not applicable for IBM

S0016 - Full testing configuration		
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0016	IBM build with intermediate heat exchanger
Rotary Valve	RV0011	RV0007 used for starting up the cooler, RV was switched when improved RV0011 ready
Valve Head	VH0028	
Measurement Station	Station 2	
Main Inlet Line	Short, Isolation, Nr. 400-000264	
Flexlines	20m, DN25, Pair 3	
Buffer Volumes	2A, 2B, 5l each, flexlines without isolator	
Compressor 1	#3, Single compressor optimization	
Compressor 2	-	
Vacuum Chamber	Chamber 2	
Lid for Vacuum Chamber	Lid 6	
Version of Software	v.56	

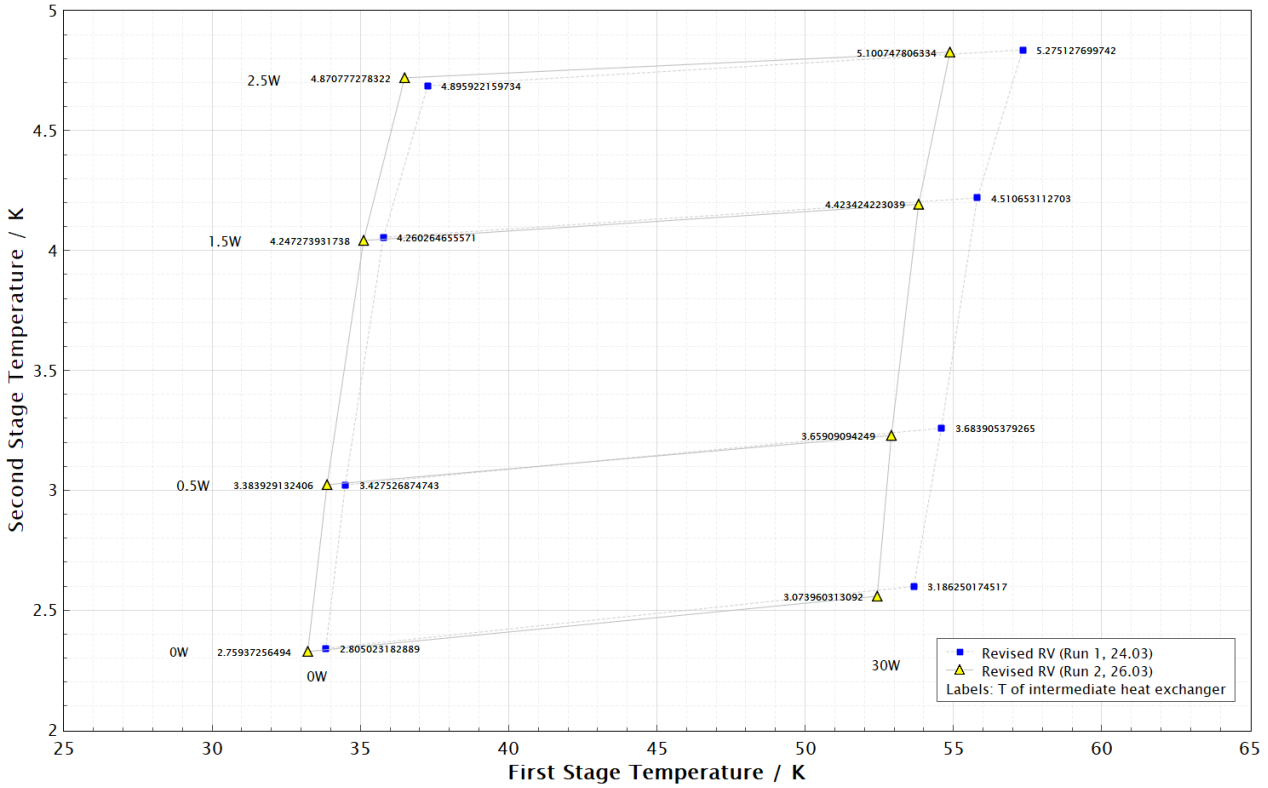
Initial filling pressure	17 bar	
Sensor T2	Cernox X232100	
Sensor T1	Cernox X235326	
Sensor T3	Cernox X231666	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm, 4 wire	
Heater Stage 1	100 Ohm, 4 wire	
Heater Stage 3	80 Ohm + Heat Switch, 2 wire	
MLI Regenerator 2	9 Layers	
MLI IHE	7 Layers	
MLI Stage 2	7.5 Layers	
MLI Regenerator 1	6.5 Layers	
MLI Pulse Tube 1	9.5 Layers	
MLI Stage 1	6.5 Layers	

S0017 - IBM PT-build #5 - 2 x F70 (shipped)

Key Information on S0017		
Cold Foot	CF0017	
Valve head	VH0029	
Rotary Valve	RV0012	
Final load maps recorded	March, 26th 2026	
Shipping date	April, 2nd 2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.5 DIV1A: 4.25	Protrusion (of fine metering valve from socket) ORV1: 11.94 mm

	DIV1B: 0.0 ORV2: 4.0 DIV2A: 8.5 DIV2B: 3.0 f / Hz: 1.1	DIV1A: 8.12 mm DIV1B: 5.23 mm ORV2: 7.56 mm DIV2A: 10.36 mm DIV2B: 6.84 mm f / Hz: 1.1
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • **Optimized for two compressors (2 x F70)**	

Load-Maps and Performance of S0017



Load maps of S0017. From the loadmaps a cooling power of appr. 33-36W @ 60K can be extracted, as is confirmed by experiments using the standard rotary valve RV0002

Key Points of the Load Map	not applicable for IBM system
Acceptance criteria	not applicable for IBM system
Control limits	not applicable for IBM system

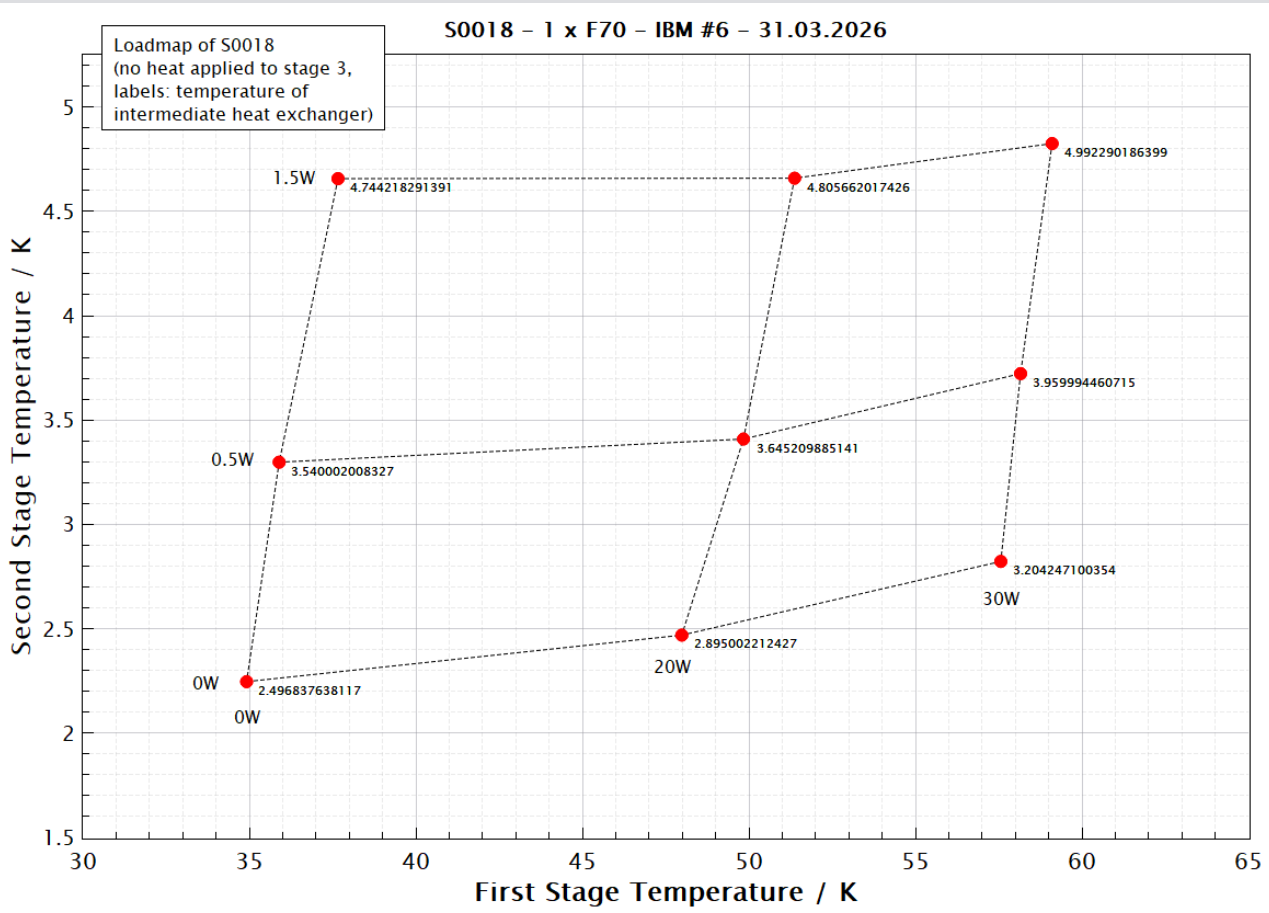
S0017 - Full testing configuration

Component	Number / Serial Number	Comment
Cold Foot	CF0017	IBM build with intermediate heat exchanger
Rotary Valve	RV0012	
Valve Head	VH0029	
Measurement Station	Station 3	
Main Inlet Line	Short, Isolation, Nr. 400-000264	
Flexlines	DN25, 20m, Pair 3	
Buffer Volumes	1A, 1B, 5l each, flexlines without isolator	
Compressor 1	#1 (F70)	double compressor optimization
Compressor 2	#2 (F70)	
Vacuum Chamber	Chamber 1	
Lid for Vacuum Chamber	Lid 4	
Version of Software	56	
Initial filling pressure	17 bar	
Sensor T2	Cernox X254360	
Sensor T1	Cernox X254340	
Sensor T3	Cernox X254356	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm, 4 wire	
Heater Stage 1	80 Ohm, 2 Wire	
Heater Stage 3	100 Ohm, 4 wire	
MLI Regenerator 2	9 Layer	
MLI IHE	8 Layer	

MLI Stage 2	7 Layer	
MLI Regenerator 1	6.5 Layers	
MLI Pulse Tube 1	9.5 Layers	
MLI Stage 1	6.5 Layers	

S0018 - IBM PT-build #6 - single F70 (shipped)

Key Information on S0018		
Cold Foot	CF0018	
Valve head	VH0030	
Rotary Valve	RV0018	
Main Inlet Line	Nr. 400-000264	
Final load maps recorded	31.03.2026	
Shipping date	13.04.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.5 DIV1A: 4.25 DIV1B: 0.5 ORV2: 4.5 DIV2A:8.625 DIV2B: 5 f / Hz: 1.0	Protrusion (of fine metering valve from socket) ORV1: 11.72 mm DIV1A: 7.79 mm DIV1B:5.27 mm ORV2: 8.09 mm DIV2A: 10.54 mm DIV2B: 8.16 mm f / Hz: 1.0
Comments / remarks	• System for IBM • MQI-PT420-aq v1.5 with brazed in heat exchanger • Optimized for a single compressor	
Load-Maps and Performance of S0018		



Key Points of the Load Map	not applicable for IBM system
Acceptance criteria	not applicable for IBM system
Control limits	not applicable for IBM system

S0018 - Full testing configuration

Component	Number / Serial Number	Comment
Cold Foot	CF0018	
Rotary Valve	RV0018	
Valve Head	VH0030	
Measurement Station	Station 1	
Main Inlet Line	Short, Isolation, Nr. 400-000264	
Flexlines	20m, DN25, Pair 3	
Buffer Volumes	3A and 3B, 5 l each, flexline with isolator	

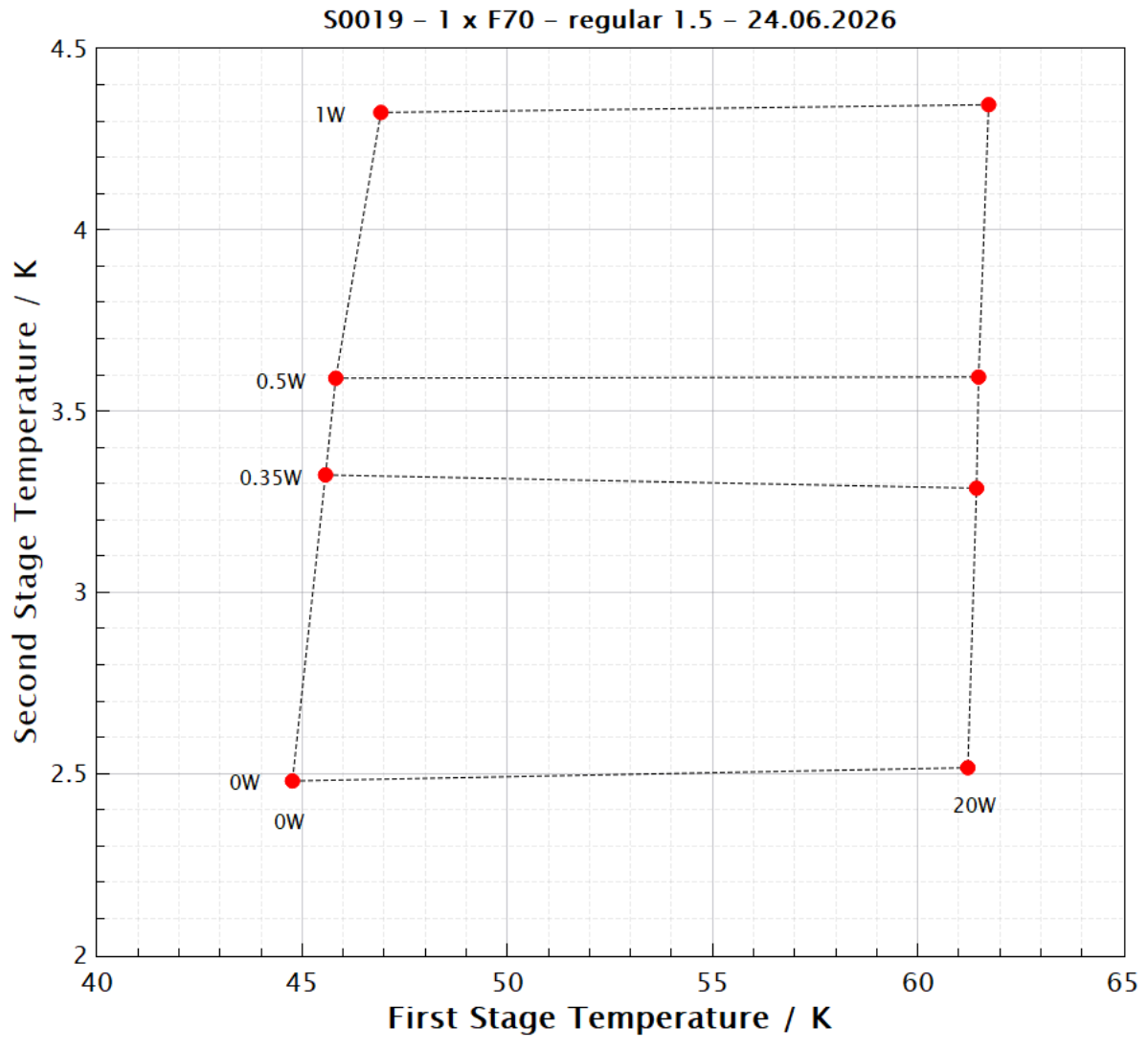
Compressor 1	#4	Single compressor tuning
Compressor 2	-	-
Vacuum Chamber	5	
Lid for Vacuum Chamber	5	
Version of Software	V56	
Initial filling pressure	17 bar	
Sensor T2	Cernox X237981	
Sensor T1	Cernox X254359	
Sensor T3	Cernox X254355	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm, 4 wire	
Heater Stage 1	47 Ohm, 2 wire	
Heater Stage 3	100 Ohm, 4 wire	
MLI Regenerator 2	10 layers	
MLI IHE	7 layers	
MLI Stage 2	7 layers	
MLI Regenerator 1	7 layers	
MLI Pulse Tube 1	10 layers	
MLI Stage 1	6 layers	

S0019 - regular 1.5 - Single F70 (shipped)

Key Information on S0019	
Cold Foot	CF0019
Valve head	VH0031
Rotary Valve	RV0017
Main Inlet Line	900-0012322/1 (Medium length)
Final load maps recorded	24.06.2026

Shipping date	31.07.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 11.5 DIV1A: 5.5 DIV1B:-0.25 ORV2:5 DIV2A:8.375 DIV2B: 4 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 12.67 mm DIV1A: 8.68 mm DIV1B: 4.31 mm ORV2: 8.22 mm DIV2A: 10.53 mm DIV2B: 8.22 mm **f / Hz: 1.1**
Comments / remarks	• Optimized for a single compressor	

Load-Maps and Performance of S0019



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 44.7, 2.48 0, 0.5, 45.8, 3.59 20, 0, 61.22, 2.51 20, 0.5, 61.48, 3.59	
Acceptance criteria	Acceptance criteria are met (version 0.1)	
Control limits	not defined upon shipping	
S0019 - Full testing configuration		
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0019	
Rotary Valve	VH0031	
Valve Head	RV0017	
Measurement Station	Station 1	
Main Inlet Line	Medium, Isolation, 900-0012322/1 (Medium length)	
Flexlines	20m, DN25, Pair 3	
Buffer Volumes	2A and 2B, 5 l each, flexline with isolator	
Compressor 1	Compressor #2	Single compressor tuning
Compressor 2	-	-
Vacuum Chamber	#3	
Lid for Vacuum Chamber	#4	
Version of Software	V57	
Initial filling pressure	17.03 bar	
Sensor T2	Cernox X254340	
Sensor T1	Cernox X254356	
Sensor T3	-	
Sensor PT2	PT100	

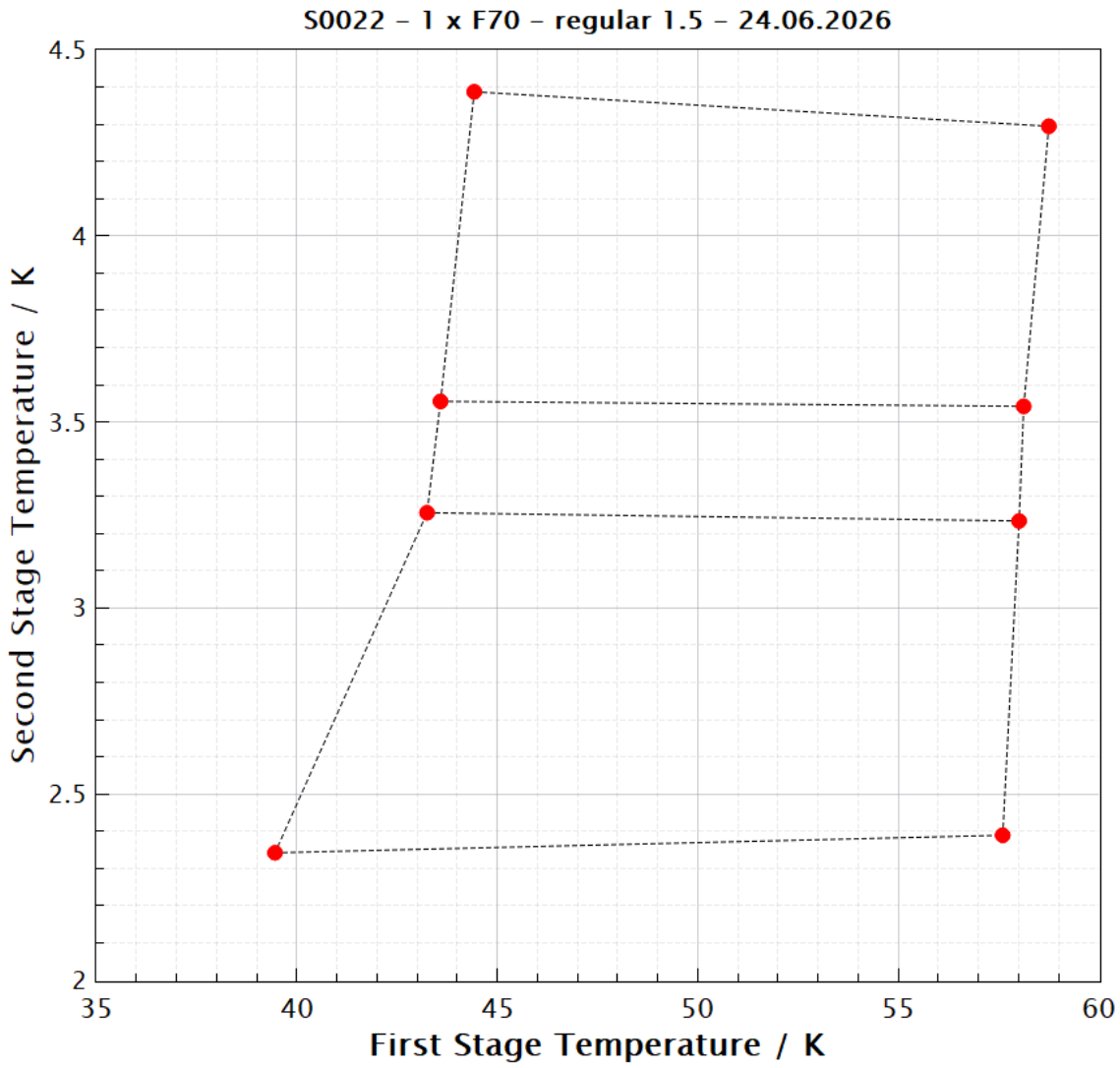
Heater Stage 2	100 Ohm	
Heater Stage 1	80 Ohm	
Heater Stage 3	-	
MLI Regenerator 2	8 layers	
MLI IHE	-	
MLI Stage 2	6 layers	
MLI Regenerator 1	6 layers	
MLI Pulse Tube 1	8 layers	
MLI Stage 1	6 layers	

S0022 - regular 1.5 - single F70 (shipped)

Key Information on S0022		
Cold Foot	CF0022	
Valve head	VH0032	
Rotary Valve	RV0019	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	24.06.2026	
Shipping date	30.06.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 9.5 DIV1A: 0.5 DIV1B: 0 ORV2: 3.75 DIV2A: 8.25 DIV2B: 5 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 11.19 mm DIV1A: 5.33 mm DIV1B: 4.64 mm ORV2: 7.28 mm DIV2A: 8.95mm DIV2B: 7.32 mm **f / Hz: 1.1**

Comments / remarks	• Optimized for a single compressor

Load-Maps and Performance of S0022



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 39.5, 2.34 0, 0.5, 43.6, 3.56 20, 0, 57.6, 2.39 20, 0.5, 58.1, 3.54
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping

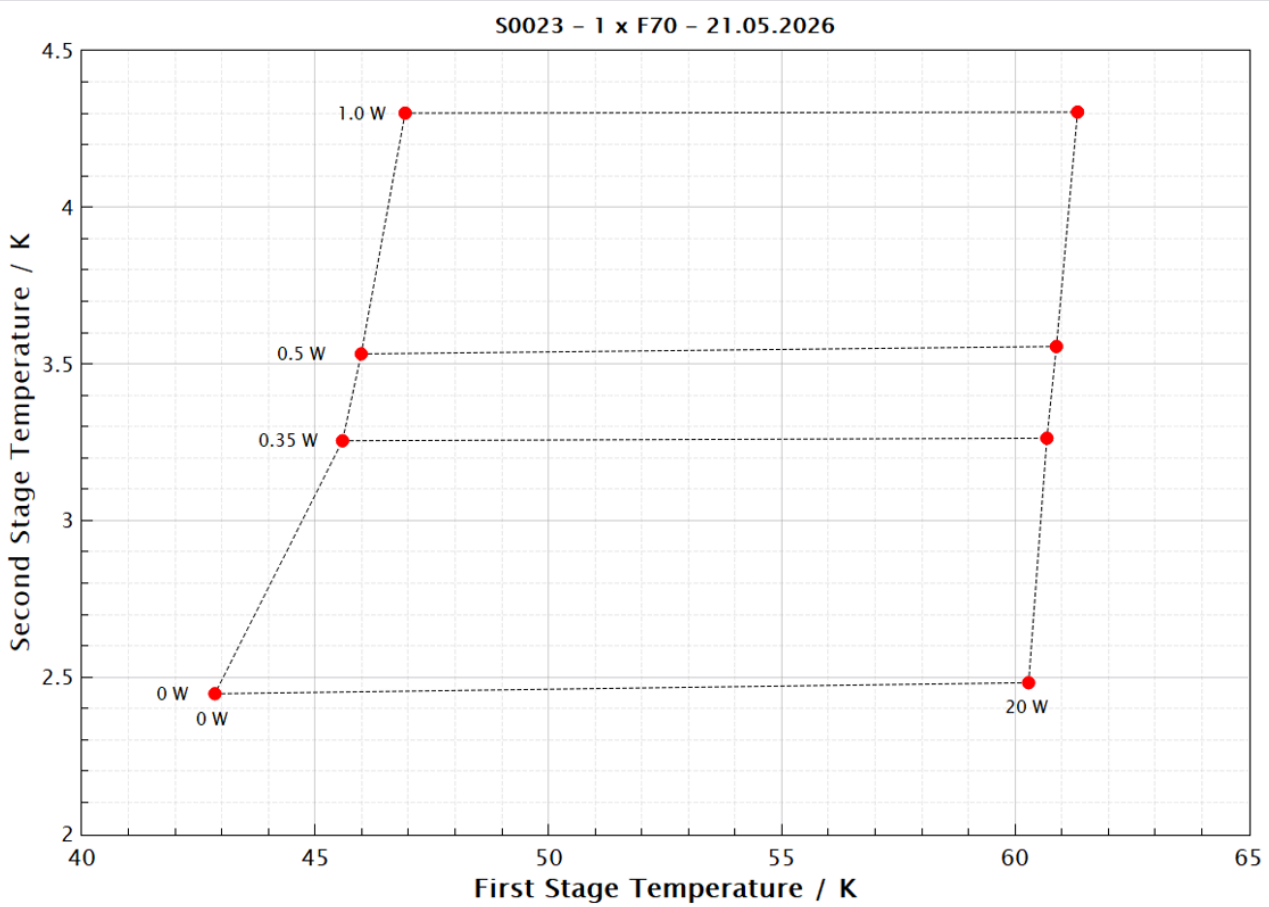
S0022 - Full testing configuration

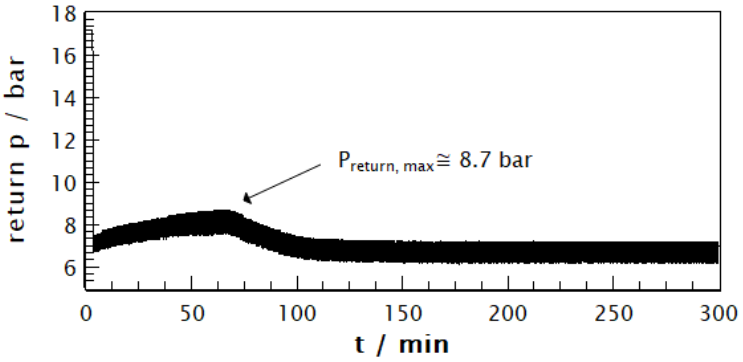
Component	Number / Serial Number	Comment
Cold Foot	CF0022	
Rotary Valve	VH0032	
Valve Head	RV0019	
Measurement Station	Station 2	
Main Inlet Line	Medium, Isolation, 900-0012322/1 (Medium length)	
Flexlines	20m, DN25, Pair 3	
Buffer Volumes	2A and 2B, 5 l each, flexline with isolator	
Compressor 1	Compressor #2	Single compressor tuning
Compressor 2	-	-
Vacuum Chamber	#5	
Lid for Vacuum Chamber	#6	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X254355	
Sensor T1	Cernox X231666	
Sensor T3	-	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm	
Heater Stage 1	80 Ohm	
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	
MLI Stage 2	10.5 layers	
MLI Regenerator 1	12 layers	

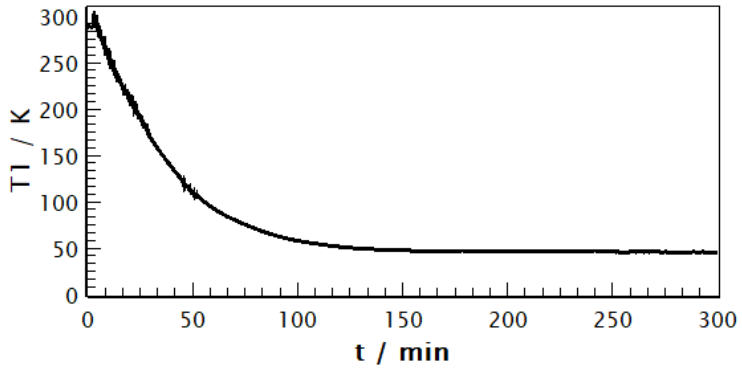
MLI Pulse Tube 1	9 layers	
MLI Stage 1	7 layers	

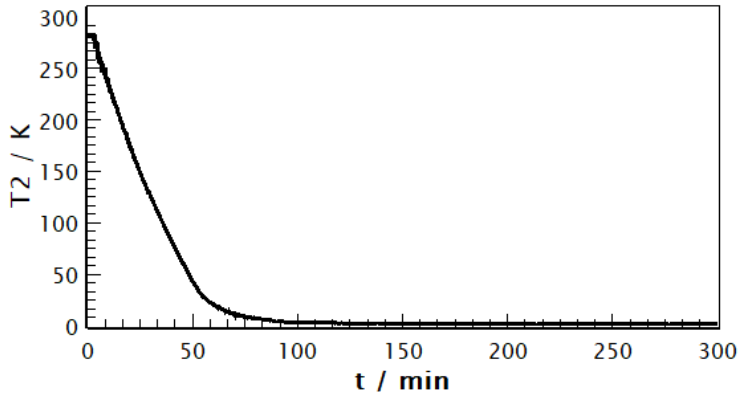
S0023 - regular 1.5 - single F70 (shipped)

Key Information on S0023		
Cold Foot	CF0023	
Valve head	VH0026	
Rotary Valve	RV0009	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	21.05.2026	
Shipping date	28.05.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.5 DIV1A: 4 DIV1B: 0.5 ORV2: 3.25 DIV2A: 8 DIV2B: 4 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 11.65 mm DIV1A: 7.5 mm DIV1B: 5.27 mm ORV2: 6.92 mm DIV2A: 9.92 mm DIV2B: 7.7 mm **f / Hz: 1.1**
Comments / remarks	• First System using a main inlet line of medium length (900-0012322/1) • Optimized for a single compressor	
Load-Maps and Performance of S0023		


S0023 – 1 x F70 – 21.05.2026
Second Stage Temperature / K
First Stage Temperature / K
4.5
4
3.5
3
2.5
2
40
45
50
55
60
65
1.0 W
0.5 W
0.35 W
0 W
0 W
20 W







Key Points of the Load Map	P1, P2, T1, T2 0, 0, 42.85, 2.45 0, 0.5, 45.99, 3.53 20, 0, 60.29, 2.4 20, 0.5, 60.88, 3.56
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping

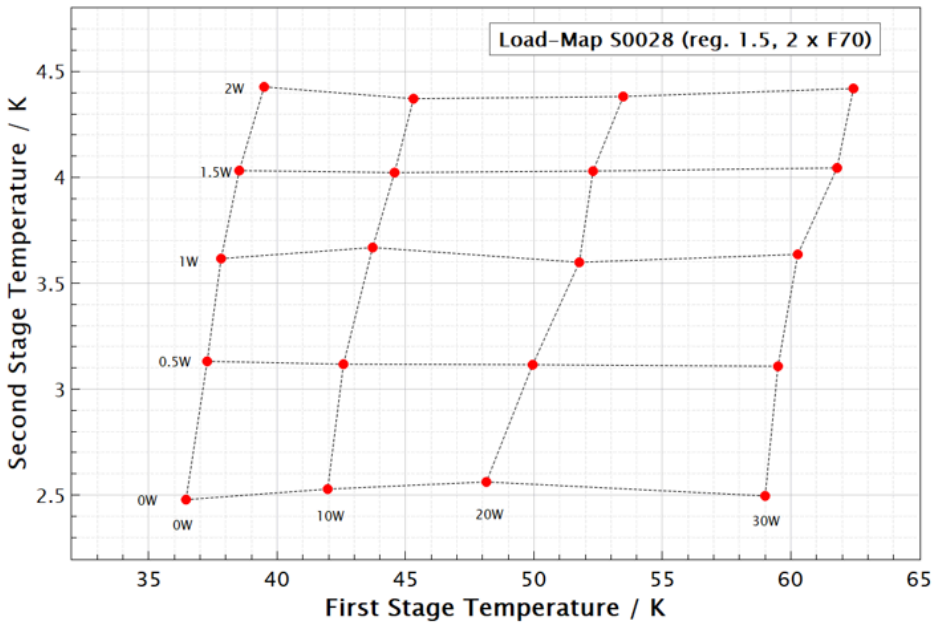
S0023 - Full testing configuration

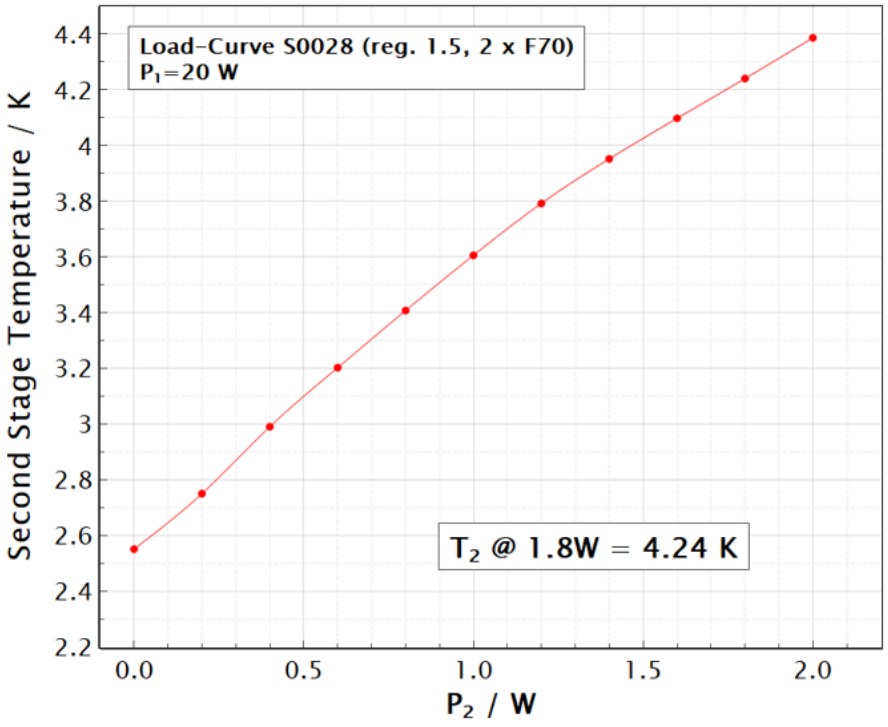
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0023	
Rotary Valve	RV0009	

Valve Head	VH0026	
Measurement Station	Station 1	
Main Inlet Line	Medium, Isolation, Nr. 900-0012322/1	
Flexlines	20m, DN25, Pair 2	
Buffer Volumes	1A and 1B, 5 l each, flexline with isolator	
Compressor 1	#4	Single compressor tuning
Compressor 2	-	-
Vacuum Chamber	#1	
Lid for Vacuum Chamber	#1	
Version of Software	V57	
Initial filling pressure	17 bar	
Sensor T2	Cernox X237981	
Sensor T1	Cernox X254359	
Sensor T3	-	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm, 4 wire	
Heater Stage 1	47 Ohm, 2 wire	
Heater Stage 3	-	
MLI Regenerator 2	12 layers	
MLI IHE	-	
MLI Stage 2	9 layers	
MLI Regenerator 1	9 layers	
MLI Pulse Tube 1	10 layers	
MLI Stage 1	8 layers	

S0028 - regular 1.5 - 2 x F70 (shipped)

Key Information on S0028		
Cold Foot	CF0028	
Valve head	VH0024	
Rotary Valve	RV0002	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	19.06.2026	
Shipping date	26.06.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 9.75 DIV1A: 4.5 DIV1B: 0.5 ORV2: 6.75 DIV2A: 8.75 DIV2B: 5.5 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 11.31 mm DIV1A: 8.03 mm DIV1B: 5.45 mm ORV2: 9.52 mm DIV2A: 10.37 mm DIV2B: 8.48 mm **f / Hz: 1.1**
Comments / remarks	• Second System using a main inlet line of medium length (900-0012322/1) • Optimized for 2 x F70	
Load-Maps and Performance of S0028		





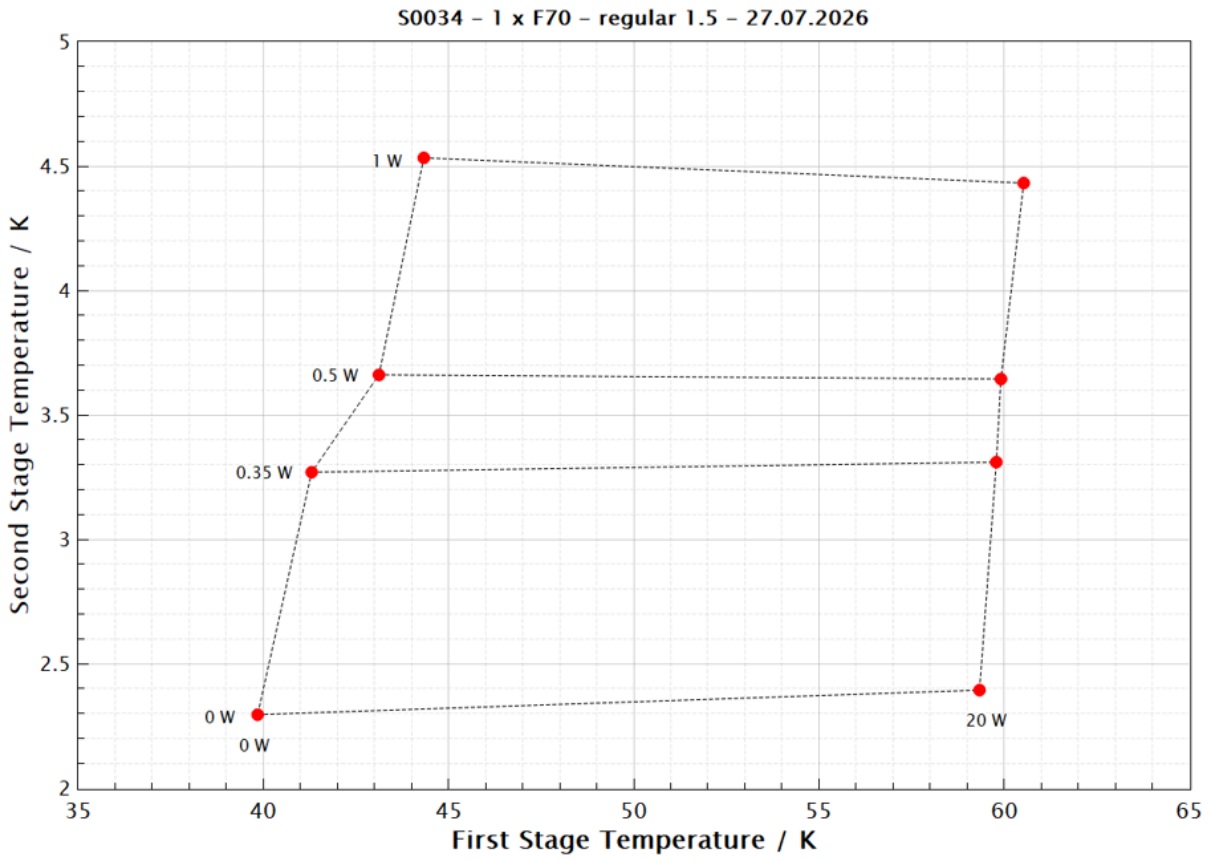
Key Points of the Load Map	P1, P2, T1, T2 0, 0, 36.5, 2.47 0, 0.5, 37.3, 3.13 20, 0, 48.1, 2.56 20, 0.5, 49.95, 3.12
Acceptance criteria	Acceptance criteria are met (version 0.1)
Control limits	not defined upon shipping
S0028 - Full testing configuration	

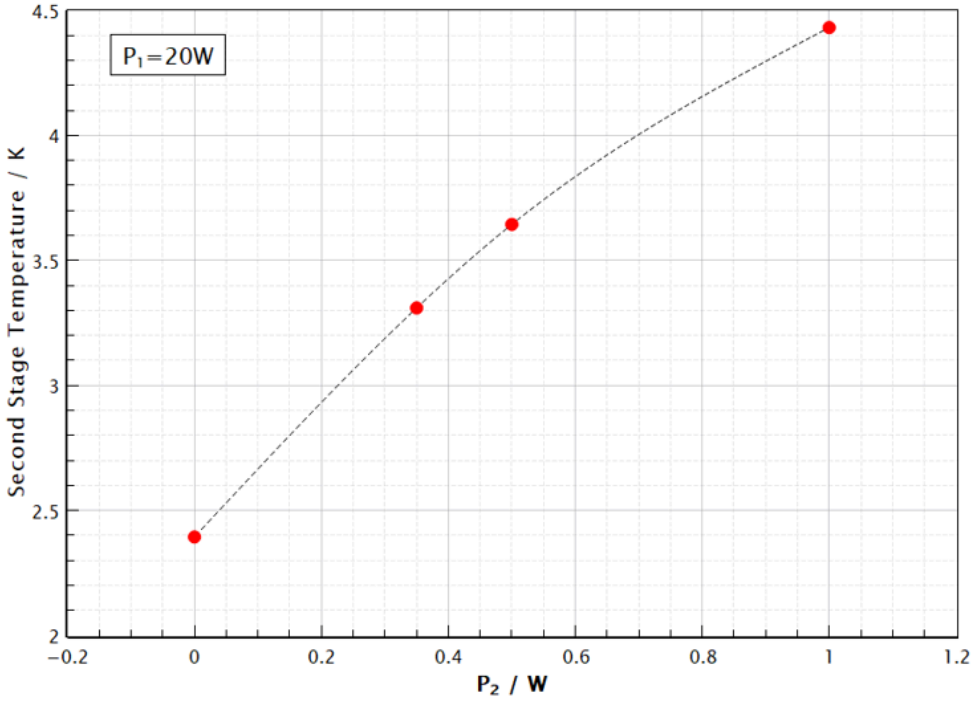
Component	Number / Serial Number	Comment
Cold Foot	CF0023	
Rotary Valve	RV0002	
Valve Head	VH0026	
Measurement Station	Station 2	
Main Inlet Line	Medium, Isolation, Nr. 900-0012322/1	
Flexlines	20m, DN25, Pair 3	
Buffer Volumes	1A and 1B, 5 l each, flexline with isolator	
Compressor 1	#1	
Compressor 2	#2	-
Vacuum Chamber	#1	
Lid for Vacuum Chamber	#2	
Version of Software	V57	
Initial filling pressure	16.95 bar	
Sensor T2	Cernox X232100	
Sensor T1	Cernox X254360	
Sensor T3	-	
Sensor PT2	PT100	
Heater Stage 2	100 Ohm, 4 wire	
Heater Stage 1	100 Ohm, 2 wire	
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	
MLI Stage 2	6.5 layers	
MLI Regenerator 1	7 layers	

MLI Pulse Tube 1	10 layers	
MLI Stage 1	8 layers	

S0034 - regular 1.5 - 1 x F70 (shipped)

Key Information on S0034		
Cold Foot	CF0034	
Valve head	RV0018	
Rotary Valve	VH0044	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	27.07.2026	
Shipping date		
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 12 DIV1A: 5.25 DIV1B: 0 ORV2: 3.125 DIV2A: 8.75 DIV2B: 5 **f / Hz: 0.9**	Protrusion (of fine metering valve from socket) ORV1: 12.44 mm DIV1A: 8.57 mm DIV1B: 4.91 mm ORV2: 7.79 mm DIV2A: 10.68 mm DIV2B: 7.97 mm **f / Hz: 0.9**
Comments / remarks		
Load-Maps and Performance of S0034		





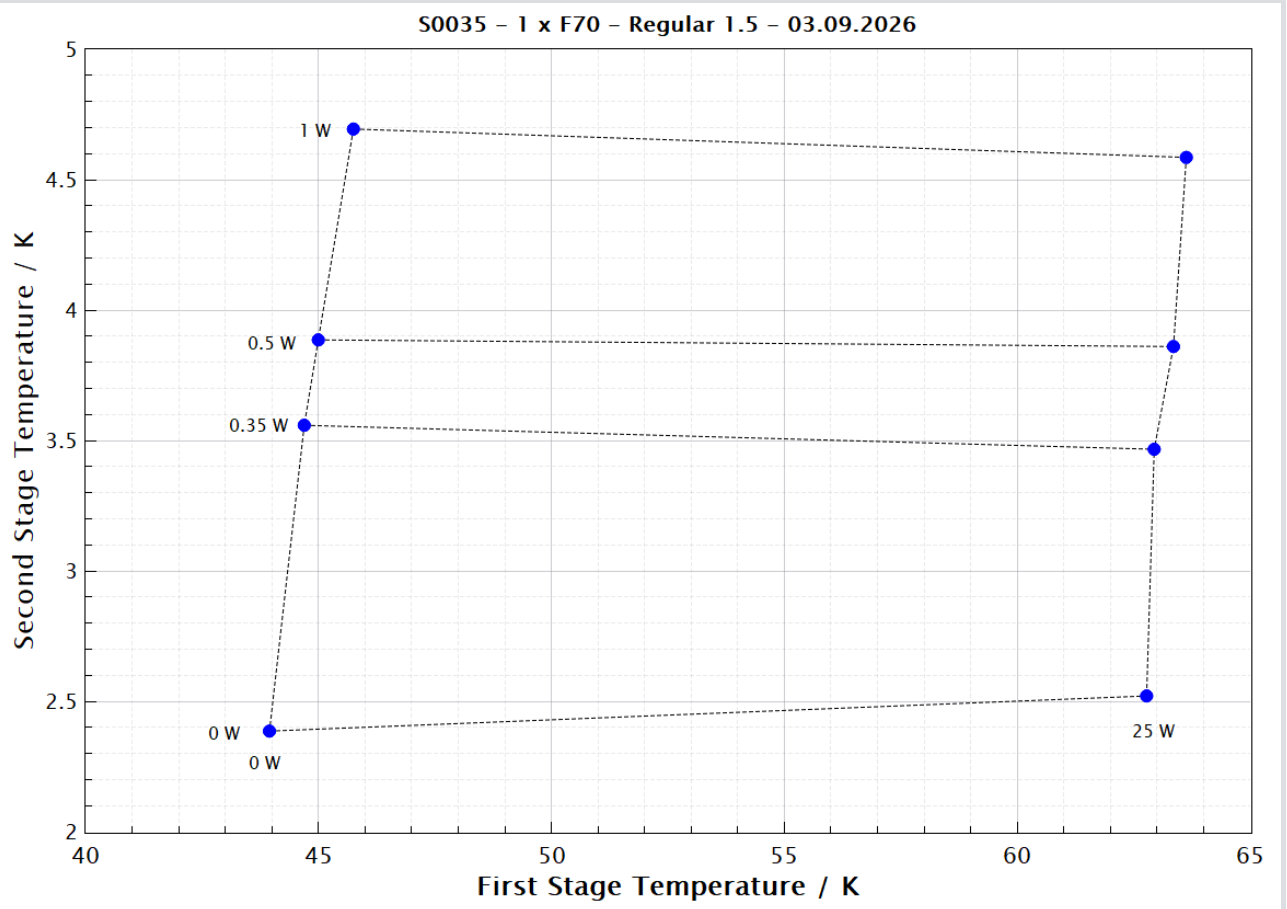
Key Points of the Load Map	P1, P2, T1, T2 0, 0, 39.8 K, 2.29 K 0, 0.5, 43.1 K, 3.66 K 20, 0, 59.3 K, 2.39 K 20, 0.5, 59.9 K, 3.64 K

Acceptance criteria	Acceptance criteria are met (version 0.1)	
Control limits	not defined upon shipping	
S0034 - Full testing configuration		
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0034	
Rotary Valve	RV0018	
Valve Head	VH0044	
Measurement Station	#3	
Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#4	
Buffer Volumes	#3	
Compressor 1	F70 #3	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#4	
Lid for Vacuum Chamber	#5	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X237981	Plug #2
Sensor T1	Cernox X254359	Plug #2
Sensor T3	-	
Sensor PT2	PT100	Plug #2
Heater Stage 2	100 Ohm	Plug #2
Heater Stage 1	80 Ohm	Plug #2
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	

MLI Stage 2	6.5 layers	
MLI Regenerator 1	7 layers	
MLI Pulse Tube 1	10 layers	
MLI Stage 1	6 layers	

S0035 - regular 1.5 - 1 x F70 (shipped)

Key Information on S0035		
Cold Foot	CF0035	
Valve head	RV0016	
Rotary Valve	VH0050	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	03.09.2026	
Shipping date		
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 12 DIV1A: 4 DIV1B: 0 ORV2: 4 DIV2A: 9 DIV2B: 7.125 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 12.82 mm DIV1A: 7.53 mm DIV1B: 5.43 mm (?) ORV2: 6.80 mm (?) DIV2A: 9.56 mm (?) DIV2B: 9.52 mm **f / Hz: 1.1** (“?“: potential problem with caliper settings inch/mm)
Comments / remarks		
Load-Maps and Performance of S0035		



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 43.95 K, 2.39 K 0, 0.5, 44.99 K, 3.88 K 25, 0, 62.78 K, 2.52 K 25, 0.5, 63.35 K, 3.86 K
Acceptance criteria	**Acceptance criteria are not met** - shipped only due to urgent need for a pulse tube
Control limits	not defined upon shipping

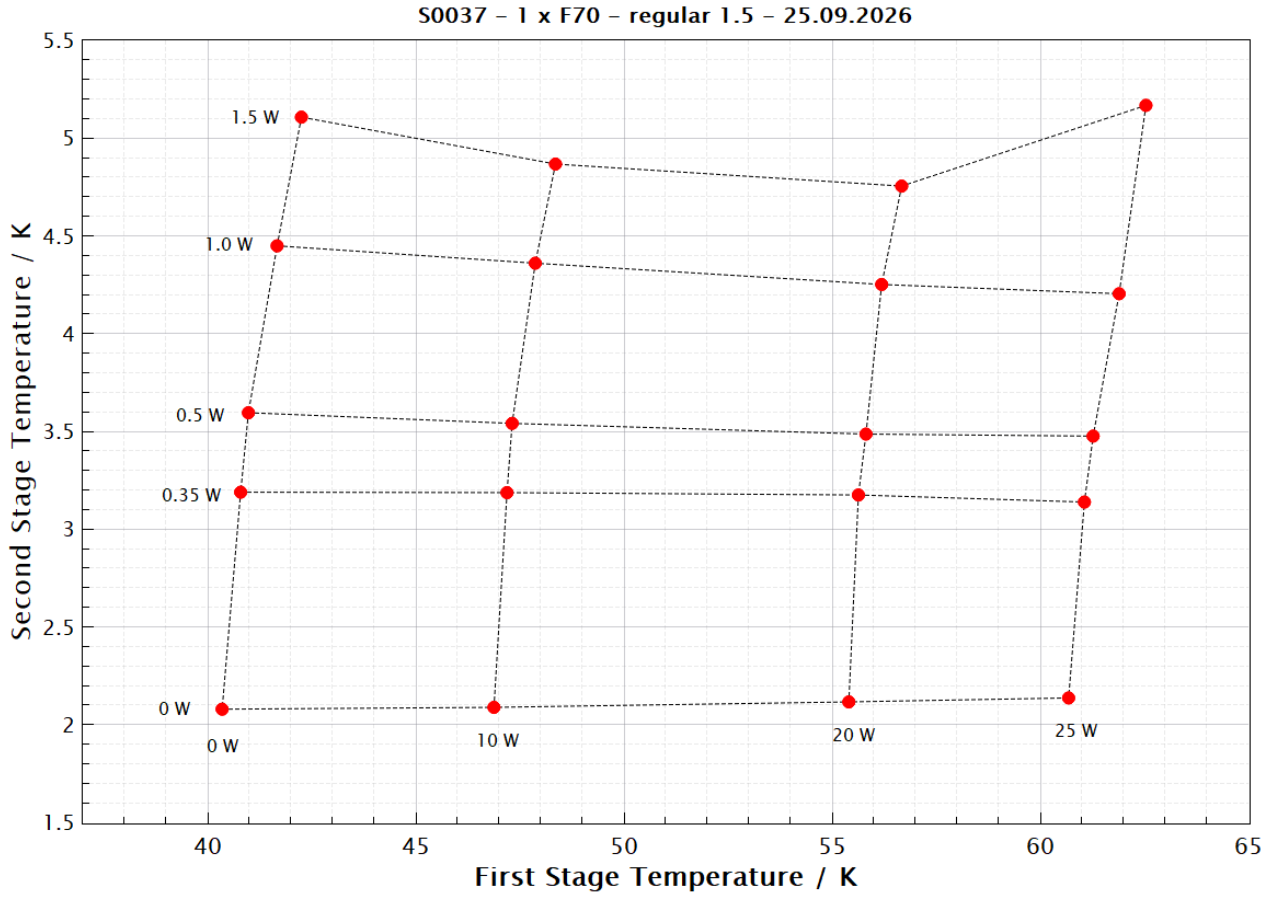
S0035 - Full testing configuration

Component	Number / Serial Number	Comment
Cold Foot	CF0035	
Rotary Valve	RV0016	
Valve Head	VH0050	
Measurement Station	#3	

Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#4	
Buffer Volumes	#3	
Compressor 1	F70 #3	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#4	
Lid for Vacuum Chamber	#6	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X256150	Plug #5
Sensor T1	Cernox X235326	Plug #5
Sensor T3	-	
Sensor PT2	PT100	Plug #5
Heater Stage 2	100 Ohm	Plug #5
Heater Stage 1	47 Ohm	Plug #5
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	
MLI Stage 2	6 layers	
MLI Regenerator 1	6 layers	
MLI Pulse Tube 1	9layers	
MLI Stage 1	6 layers	

S0037 - regular 1.5 - 1 x F70 (ready to be shipped)

Key Information on S0037		
Cold Foot	CF0037	
Valve head	RV0026	
Rotary Valve	VH0022	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	25.09.2026	
Shipping date	not shipped yet	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 11.5 DIV1A: 3.25 DIV1B: 0.75 ORV2: 2.875 DIV2A: 9 DIV2B: 4.25 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 12.25 mm DIV1A: 6.39 mm DIV1B: 5.55 mm ORV2: 6.86 mm DIV2A: 10.78 mm DIV2B: 7.5 mm **f / Hz: 1.1**
Comments / remarks		
Load-Maps and Performance of S0037		



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 40.34, 2.08 0, 0.5, 40.98, 3.60 25, 0, 60.68, 2.14 25, 0.5, 61.27, 3.48
Acceptance criteria	Modified Acceptance Criteria: 3.5K @ 0.5W and 60K @ 25W
Control limits	not defined upon shipping

S0037 - Full testing configuration

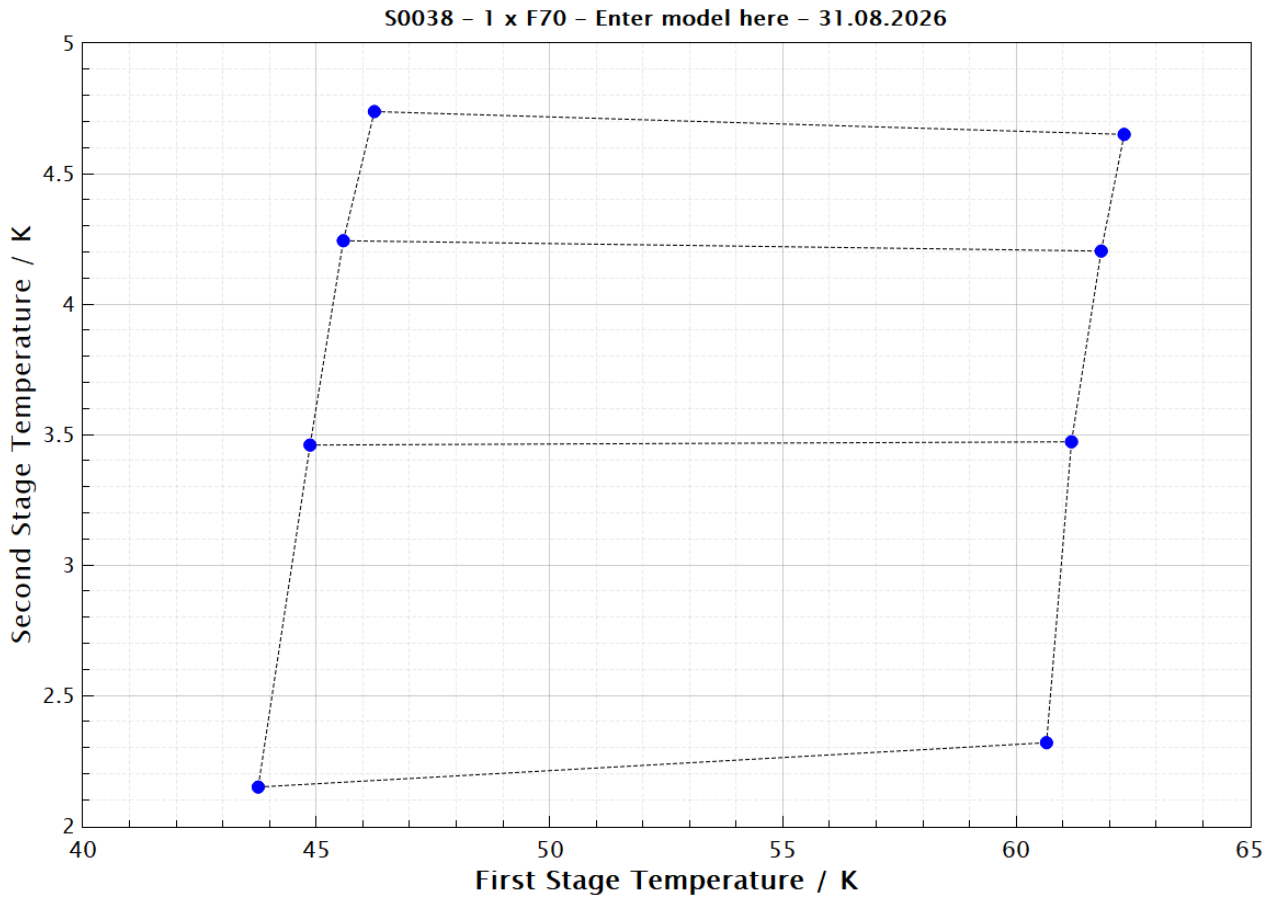
Component	Number / Serial Number	Comment
Cold Foot	CF0037	
Rotary Valve	RV0022	
Valve Head	VH0026	
Measurement Station	#1	
Main Inlet Line	900-0012322/1 (Medium length)	

Flexlines	#2	
Buffer Volumes	#1	
Compressor 1	F70 #3	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#2	
Lid for Vacuum Chamber	#1	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X254355	Plug #3
Sensor T1	Cernox X231666	Plug #3
Sensor T3	-	
Sensor PT2	PT100	Plug #3
Heater Stage 2	100 Ohm	Plug #3
Heater Stage 1	47 Ohm	Plug #3
Heater Stage 3	-	
MLI Regenerator 2	10.5 layers	
MLI IHE	-	
MLI Stage 2	6.5 layers	
MLI Regenerator 1	7.5 layers	
MLI Pulse Tube 1	12 layers	
MLI Stage 1	8 layers	

S0038 - regular 1.5 - 1 x F70 (shipped)

Key Information on S0038

Cold Foot	CF0038	
Valve head	RV0007	
Rotary Valve	VH0038	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	31.08.2026	
Shipping date	not shipped yet	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 12.25 DIV1A: 7.5 DIV1B: 0 ORV2: 3.5 DIV2A: 8.5 DIV2B: 7.25 **f / Hz: 1.0**	Protrusion (of fine metering valve from socket) ORV1: 13.17 mm DIV1A: 9.61 mm DIV1B: 4.67 mm ORV2: 7.29 mm DIV2A:10.24 mm DIV2B: 9.51 mm **f / Hz: 1.0**
Comments / remarks		
Load-Maps and Performance of S0038		



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 43.8, 2.15 0, 0.5, 44.86, 3.46 25, 0, 60.64, 2.31 25, 0.5, 61.18, 3.47
Acceptance criteria	Modified Acceptance Criteria: 3.5K @ 0.5W and 60K @ 25W
Control limits	not defined upon shipping

S0038 - Full testing configuration

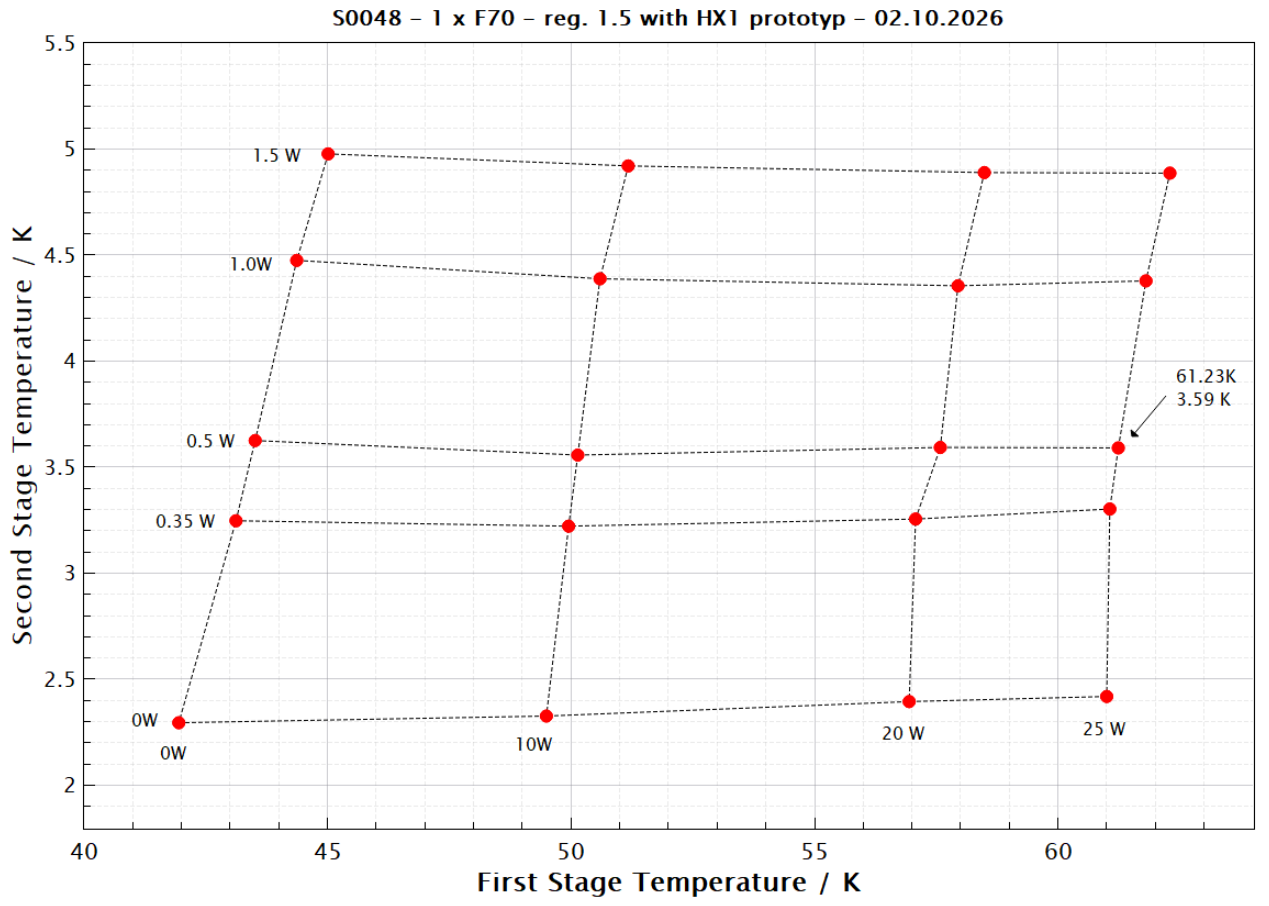
Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0038	
Rotary Valve	RV0007	
Valve Head	VH0038	
Measurement Station	#1	

Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#2	
Buffer Volumes	#1	
Compressor 1	F70 #4	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#2	
Lid for Vacuum Chamber	#1	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2	Cernox X254355	Plug #3
Sensor T1	Cernox X231666	Plug #3
Sensor T3	-	
Sensor PT2	PT100	Plug #3
Heater Stage 2	100 Ohm	Plug #3
Heater Stage 1	47 Ohm	Plug #3
Heater Stage 3	-	
MLI Regenerator 2	9 layers	
MLI IHE	-	
MLI Stage 2	8 layers	
MLI Regenerator 1	7 layers	
MLI Pulse Tube 1	7 layers	
MLI Stage 1	6 layers	

S0048 - regular 1.5 with prototype 1st stage heat exchanger - 1 x F70

Key Information on S0048

Cold Foot	CF0048	
Valve head	RV0024	
Rotary Valve	VH0023	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	02.10.2026	
Shipping date	12.10.2026	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10.25 DIV1A: 4 DIV1B: 0 ORV2: 3.5 DIV2A: 9.5 DIV2B: 5.75 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 11.33 mm DIV1A: 7.64 mm DIV1B: 4.99 mm ORV2: 7.39 mm DIV2A: 10.56 mm DIV2B: 8.72 mm **f / Hz: 1.1**
Comments / remarks	modified first stage heat exchanger (Cu block with tubes)	
Load-Maps and Performance of S0048		



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 41.94, 2.29 0, 0.5, 43.51, 3.62 25, 0, 61.00, 2.41 25, 0.5, 61.23, 2.59
Acceptance criteria	Modified Acceptance Criteria: 3.5K @ 0.5W and 60K @ 25W
Control limits	not defined upon shipping

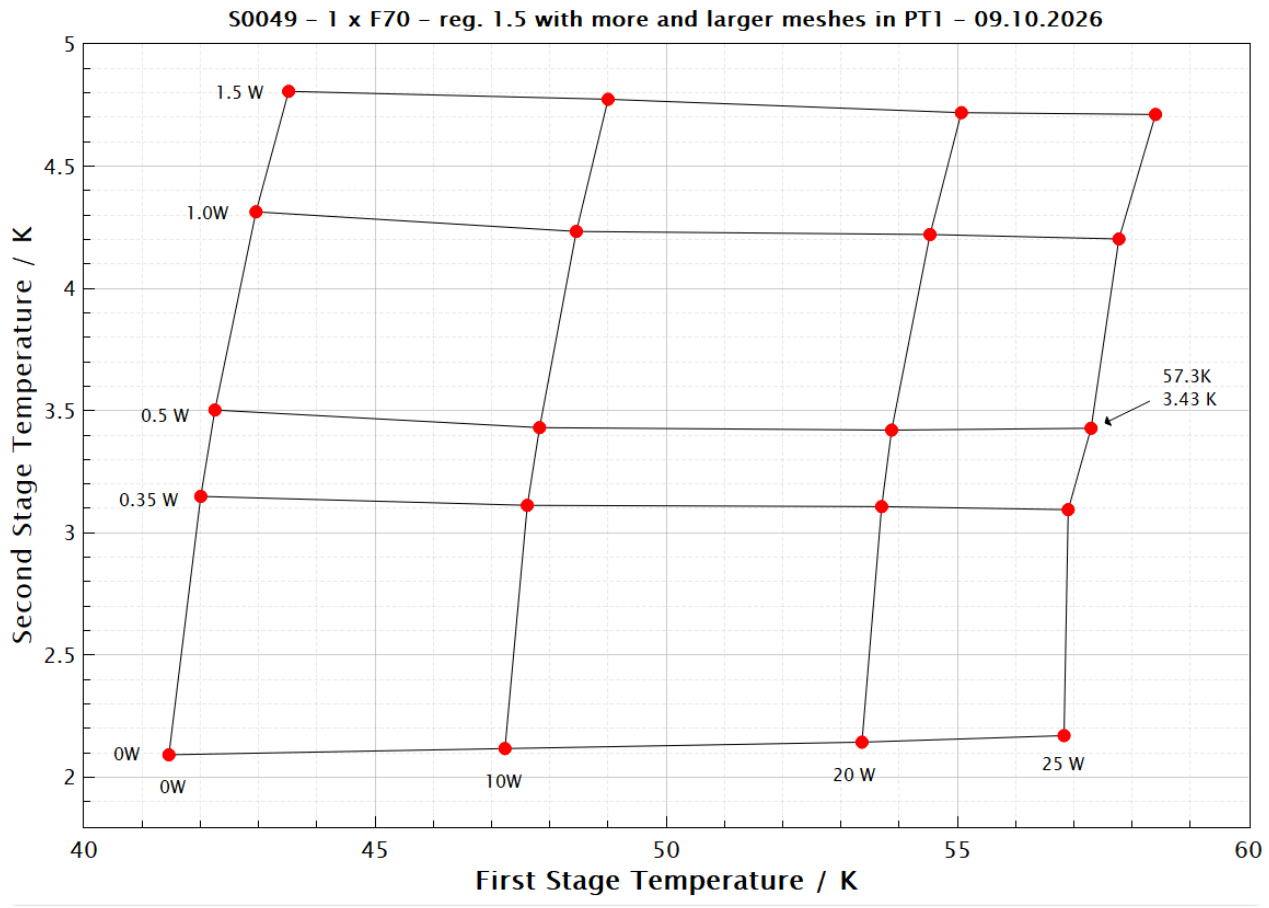
S0048 - Full testing configuration

Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0048	
Rotary Valve	RV0024	
Valve Head	VH0023	
Measurement Station	#3	

Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#4	
Buffer Volumes	#3	
Compressor 1	F70 #4	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#4	
Lid for Vacuum Chamber	#6	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2		Plug #5
Sensor T1		Plug #5
Sensor T3	-	
Sensor PT2	PT100	Plug #5
Heater Stage 2	100 Ohm	Plug #5
Heater Stage 1	80 Ohm	Plug #5
Heater Stage 3	-	
MLI Regenerator 2	10 layers	
MLI IHE	-	
MLI Stage 2	7 layers	
MLI Regenerator 1	8 layers	
MLI Pulse Tube 1	9 layers	
MLI Stage 1	6 layers	

S0049 - regular 1.5 with prototype 1st stage heat exchanger - 1 x F70

Key Information on S0049		
Cold Foot	CF0049	
Valve head	RV0030	
Rotary Valve	VH0059	
Main Inlet Line	900-0012322/1 (Medium length)	
Final load maps recorded	09.10.2026	
Shipping date	not shipped yet	
Final valve settings:	Rotations (counted relative to fully closed valve) ORV1: 10 DIV1A: 4 DIV1B: -0.25 ORV2: 4.125 DIV2A: 9.5 DIV2B: 6.125 **f / Hz: 1.1**	Protrusion (of fine metering valve from socket) ORV1: 11.36 mm DIV1A: 7.42 mm DIV1B: 4.75 mm ORV2: 7.56 mm DIV2A: 11.35 mm DIV2B: 9.29 mm **f / Hz: 1.1**
Comments / remarks	modified first stage heat exchanger (Cu block with tubes)	
Load-Maps and Performance of S0049		



Key Points of the Load Map	P1, P2, T1, T2 0, 0, 41.5, 2.09 0, 0.5, 42.3, 3.50 25, 0, 56.8, 2.17 25, 0.5, 57.3, 3.43
Acceptance criteria	Modified Acceptance Criteria: 3.5K @ 0.5W and 60K @ 25W
Control limits	not defined upon shipping

S0049 - Full testing configuration

Component	**Number / Serial Number**	**Comment**
Cold Foot	CF0049	
Rotary Valve	RV0030	
Valve Head	VH0059	
Measurement Station	#3	

Main Inlet Line	900-0012322/1 (Medium length)	
Flexlines	#4	
Buffer Volumes	#3	
Compressor 1	F70 #4	single compressor tuning
Compressor 2	-	
Vacuum Chamber	#4	
Lid for Vacuum Chamber	#6	
Version of Software	V57	
Initial filling pressure	17.1 bar	
Sensor T2		Plug #3
Sensor T1		Plug #3
Sensor T3	-	
Sensor PT2	PT100	Plug #3
Heater Stage 2	100 Ohm	Plug #3
Heater Stage 1	80 Ohm	Plug #3
Heater Stage 3	-	
MLI Regenerator 2	9layers	
MLI IHE	-	
MLI Stage 2	6 layers	
MLI Regenerator 1	9 layers	
MLI Pulse Tube 1	10 layers	
MLI Stage 1	7 layers	